

# Sid Mathur's Thematic ETF Screener – Daily Report

2026-06-03

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## Action Center

**Run date:** 2026-06-03    **Universe:** 660 ETFs / 26 clusters

**Full PC1:** LONG since 2026-04-13 (D+35, 5.7%)    **Clusters long:** 14/26    **ETFs long:** 362/660 (55%)

### Today's events (cluster level)

$\text{spread\_buy} = 5\text{dma} - 50\text{dma}$ ;  $\text{spread\_sell} = 5\text{dma} - 20\text{dma}$  (price units). A BUY fired today iff  $\text{spread\_buy}$  crossed above 0; a SELL iff  $\text{spread\_sell}$  crossed below the cushioned threshold.

*No new cluster crossovers today.*

### Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p\_h5 is the 5-day probability for context.  **$\text{spread\_buy} = 5\text{dma} - 50\text{dma}$ ,  $\text{spread\_sell} = 5\text{dma} - 20\text{dma}$**  (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

#### Imminent BUY (top 5 by h=1 NSS, filter-approved)

| cluster    | theme                           | spread_buy | p_h1 | p_h5 | nss_h1 | nss_h5 |
|------------|---------------------------------|------------|------|------|--------|--------|
| cluster_2  | U.S. Mid Cap Equity             | 5.699      | 1    | 1.00 | 1.000  | 0.99   |
| cluster_22 | Japan Equity                    | 8.677      | 1    | 1.00 | 1.000  | 0.99   |
| cluster_10 | High Yield Bond                 | 0.754      | 1    | 0.99 | 0.943  | 0.77   |
| cluster_9  | Materials                       | 2.459      | 1    | 0.96 | 0.767  | 0.64   |
| cluster_24 | Cash / Ultra-Short Fixed Income | 0.236      | 1    | 1.00 | 0.699  | 0.38   |

#### Imminent SELL (top 2 by h=1 NSS)

| cluster    | theme                            | spread_sell | p_h1  | p_h5 | nss_h1 | nss_h5 |
|------------|----------------------------------|-------------|-------|------|--------|--------|
| cluster_25 | International Fixed Income       | -0.129      | 0.858 | 0.72 | -0.259 | -0.21  |
| cluster_21 | Inflation-Protected Fixed Income | -0.096      | 0.919 | 0.84 | -0.089 | -0.36  |

## Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

**Replication basket:** VOO basket cor: 0.94 sticky

**Current state:** LONG since 2026-04-13 (35 days, 5.7%)

5dma: 142.631 20dma: 140.832 50dma: 136.862

Spread vs 50dma: +5.769 Spread vs 20dma: +1.799

P(buy crossover within 5 days): 100.0% P(sell): 0.4%

## Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

### Active positions (long clusters)

| ticker | cluster    | theme                          | nss    | days | return | pos_wt | cor  |
|--------|------------|--------------------------------|--------|------|--------|--------|------|
| VOO    | cluster_1  | U.S. Large Cap Equity          | 1.000  | 35   | 16.4%  | 5.80%  | 0.98 |
| VT     | cluster_5  | Global Equity                  | 1.000  | 35   | 13.2%  | 5.64%  | 0.94 |
| VTV    | cluster_3  | U.S. Large Cap Value           | 0.999  | 35   | 3.8%   | 8.61%  | 0.98 |
| VHT    | cluster_11 | Healthcare                     | 0.999  | 18   | -1.3%  | 5.44%  | 0.93 |
| VEA    | cluster_4  | Developed ex-US Equity         | 0.989  | 36   | 3.4%   | 4.26%  | 0.99 |
| IEMG   | cluster_7  | Emerging Markets Equity        | 0.987  | 36   | 13.1%  | 3.31%  | 0.99 |
| MUB    | cluster_16 | Municipal Bond                 | 0.970  | 3    | 0.1%   | 18.09% | 0.95 |
| TLT    | cluster_13 | Fixed Income                   | 0.854  | 2    | -0.2%  | 7.33%  | 0.97 |
| BND    | cluster_6  | Aggregate Fixed Income         | 0.577  | 2    | -0.4%  | 14.57% | 0.99 |
| SRLN   | cluster_23 | Bank Loan / Floating Rate      | 0.545  | 39   | 1.1%   | 6.15%  | 0.89 |
| FLOT   | cluster_23 | Bank Loan / Floating Rate      | 0.545  | 39   | 1.1%   | 5.87%  | 0.89 |
| BKLN   | cluster_23 | Bank Loan / Floating Rate      | 0.545  | 39   | 1.1%   | 4.23%  | 0.89 |
| FLRN   | cluster_23 | Bank Loan / Floating Rate      | 0.545  | 39   | 1.1%   | 2.52%  | 0.89 |
| FLTR   | cluster_23 | Bank Loan / Floating Rate      | 0.545  | 39   | 1.1%   | 0.70%  | 0.89 |
| BBAX   | cluster_8  | Asia-Pacific Equity            | 0.157  | 6    | 0.3%   | 2.55%  | 0.97 |
| IGF    | cluster_8  | Asia-Pacific Equity            | 0.157  | 6    | 0.3%   | 1.63%  | 0.97 |
| EWZ    | cluster_8  | Asia-Pacific Equity            | 0.157  | 6    | 0.3%   | 0.53%  | 0.97 |
| VNQ    | cluster_15 | Real Estate                    | 0.111  | 36   | 1.9%   | 5.66%  | 0.99 |
| BWX    | cluster_25 | International Fixed Income     | -0.209 | 2    | -0.5%  | 9.12%  | 0.93 |
| VTIP   | cluster_21 | Inflation-Protected Fixed I... | -0.359 | 4    | -0.7%  | 34.46% | 0.90 |

### Watchlist (imminent-buy clusters, $p > 80\%$ , not yet long)

| ticker | cluster    | theme                          | p_h5 | nss   | weight | cor  |
|--------|------------|--------------------------------|------|-------|--------|------|
| IJH    | cluster_2  | U.S. Mid Cap Equity            | 1.00 | 0.992 | 1.020  | 0.99 |
| EWJ    | cluster_22 | Japan Equity                   | 1.00 | 0.987 | 0.908  | 0.96 |
| ICSH   | cluster_24 | Cash / Ultra-Short Fixed In... | 1.00 | 0.376 | 0.406  | 0.84 |
| SHV    | cluster_24 | Cash / Ultra-Short Fixed In... | 1.00 | 0.376 | 0.298  | 0.84 |
| MINT   | cluster_24 | Cash / Ultra-Short Fixed In... | 1.00 | 0.376 | 0.179  | 0.84 |
| GBIL   | cluster_24 | Cash / Ultra-Short Fixed In... | 1.00 | 0.376 | 0.122  | 0.84 |

|      |            |                 |      |       |       |      |
|------|------------|-----------------|------|-------|-------|------|
| HYG  | cluster_10 | High Yield Bond | 0.99 | 0.771 | 0.916 | 0.95 |
| GUNR | cluster_9  | Materials       | 0.96 | 0.635 | 0.773 | 0.96 |
| AML  | cluster_9  | Materials       | 0.96 | 0.635 | 0.114 | 0.96 |
| URA  | cluster_9  | Materials       | 0.96 | 0.635 | 0.081 | 0.96 |
| XLF  | cluster_14 | Financials      | 0.88 | 0.367 | 1.097 | 0.96 |

## Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

|    | id         | name                           | category             | n  | signal | change  | days | return | nss    | dNSS   |
|----|------------|--------------------------------|----------------------|----|--------|---------|------|--------|--------|--------|
| 1  | cluster_1  | U.S. Large Cap Equity          | U.S. Large Cap Eq... | 84 | LONG   | shifted | 35   | 16.4%  | 1.000  | +0.020 |
| 5  | cluster_5  | Global Equity                  | Global Equity        | 49 | LONG   | shifted | 35   | 13.2%  | 1.000  | +0.288 |
| 3  | cluster_3  | U.S. Large Cap Value           | U.S. Large Cap Va... | 77 | LONG   | shifted | 35   | 3.8%   | 0.999  | +0.002 |
| 11 | cluster_11 | Healthcare                     | Healthcare           | 17 | LONG   | shifted | 18   | -1.3%  | 0.999  | +0.275 |
| 2  | cluster_2  | U.S. Mid Cap Equity            | U.S. Mid Cap Equi... | 79 | FLAT   | shifted |      |        | 0.992  | -0.008 |
| 4  | cluster_4  | Developed ex-US Equity         | Developed ex-US E... | 61 | LONG   | shifted | 36   | 3.4%   | 0.989  | -0.010 |
| 7  | cluster_7  | Emerging Markets Equity        | Emerging Markets ... | 25 | LONG   | shifted | 36   | 13.1%  | 0.987  | -0.013 |
| 22 | cluster_22 | Japan Equity                   | Japan Equity         | 9  | FLAT   |         |      |        | 0.987  | -0.007 |
| 16 | cluster_16 | Municipal Bond                 | Municipal Bond       | 13 | LONG   |         | 3    | 0.1%   | 0.970  | +0.062 |
| 13 | cluster_13 | Fixed Income                   | Fixed Income         | 15 | LONG   | shifted | 2    | -0.2%  | 0.854  | +1.658 |
| 10 | cluster_10 | High Yield Bond                | High Yield Bond      | 21 | FLAT   | shifted |      |        | 0.771  | +0.244 |
| 9  | cluster_9  | Materials                      | Materials            | 21 | FLAT   | shifted |      |        | 0.635  | +0.336 |
| 6  | cluster_6  | Aggregate Fixed Income         | Aggregate Fixed I... | 43 | LONG   | shifted | 2    | -0.4%  | 0.577  | -0.423 |
| 23 | cluster_23 | Bank Loan / Floating Rate      | Bank Loan / Float... | 9  | LONG   |         | 39   | 1.1%   | 0.545  | -0.345 |
| 24 | cluster_24 | Cash / Ultra-Short Fixed In... | Cash / Ultra-Shor... | 9  | FLAT   |         |      |        | 0.376  | -0.553 |
| 14 | cluster_14 | Financials                     | Financials           | 14 | FLAT   |         |      |        | 0.367  | -0.176 |
| 8  | cluster_8  | Asia-Pacific Equity            | Asia-Pacific Equi... | 23 | LONG   | shifted | 6    | 0.3%   | 0.157  | -0.821 |
| 15 | cluster_15 | Real Estate                    | Real Estate          | 14 | LONG   |         | 36   | 1.9%   | 0.111  | -0.353 |
| 25 | cluster_25 | International Fixed Income     | International Fix... | 6  | LONG   |         | 2    | -0.5%  | -0.209 | -0.019 |
| 21 | cluster_21 | Inflation-Protected Fixed I... | Inflation-Protect... | 10 | LONG   |         | 4    | -0.7%  | -0.359 | -0.473 |
| 17 | cluster_17 | Gold                           | Gold                 | 12 | FLAT   |         |      |        | -0.512 | -0.019 |
| 12 | cluster_12 | China Equity                   | China Equity         | 15 | FLAT   | shifted |      |        | -0.782 | -1.782 |
| 20 | cluster_20 | Broad Commodities              | Broad Commodities    | 10 | FLAT   |         |      |        | -0.987 | -0.008 |
| 18 | cluster_18 | Consumer Staples               | Consumer Staples     | 11 | FLAT   |         |      |        | -0.999 | -0.025 |
| 26 | cluster_26 | Leveraged / Inverse            | Leveraged / Inver... | 2  | FLAT   |         |      |        | -0.999 | -0.001 |
| 19 | cluster_19 | India Equity                   | India Equity         | 11 | FLAT   |         |      |        | -1.000 | -0.000 |

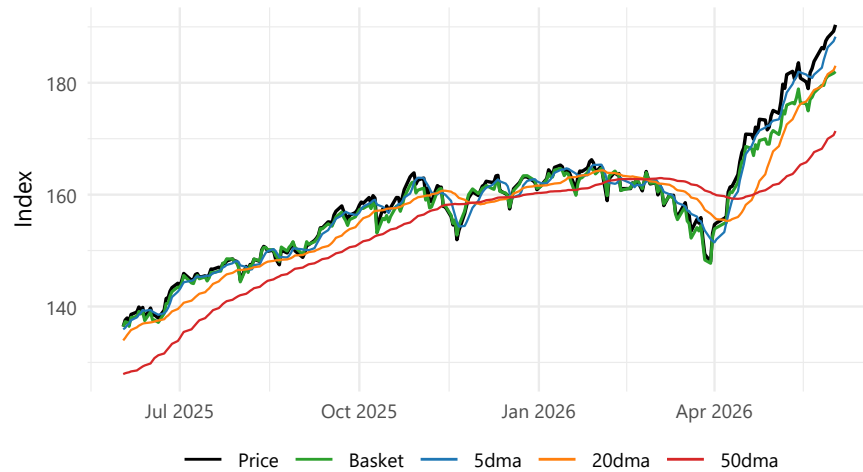
## Cluster Details

### Cluster 1 – U.S. Large Cap Equity

**Members:** 84   **Var explained:** 0.895   **Signal:** LONG since 2026-04-13 (35 days, 16.4%)

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +32.2%)
- Latest sell\_cross: 2026-03-06
- Current 252d return: +40.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



#### Category breakdown

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| Technology                   | 19        | 0.2187     |
| U.S. Large Cap Growth        | 11        | 0.1358     |
| U.S. Large Cap Equity        | 9         | 0.1121     |
| U.S. Total Market Equity     | 7         | 0.0873     |
| Consumer Discretionary       | 5         | 0.0557     |
| U.S. Equity                  | 3         | 0.0373     |
| U.S. Mega Cap Equity         | 3         | 0.0372     |
| U.S. Quality / Factor Equity | 3         | 0.0360     |

|                             |   |        |
|-----------------------------|---|--------|
| U.S. Multifactor Equity     | 3 | 0.0358 |
| Communication Services      | 3 | 0.0327 |
| U.S. ESG Equity             | 2 | 0.0249 |
| Growth Equity               | 2 | 0.0243 |
| Global Equity               | 2 | 0.0242 |
| Large Cap Blend Equities    | 1 | 0.0124 |
| Commodity                   | 1 | 0.0124 |
| U.S. Mega Cap Growth        | 1 | 0.0123 |
| Large Cap Value Equities    | 1 | 0.0122 |
| U.S. Equal Weight Equity    | 1 | 0.0120 |
| Dividend Equity             | 1 | 0.0119 |
| U.S. Dividend Equity        | 1 | 0.0116 |
| Options Income              | 1 | 0.0112 |
| U.S. Moat Equity            | 1 | 0.0111 |
| U.S. Small Cap Technology   | 1 | 0.0108 |
| Trend-Following Equity      | 1 | 0.0106 |
| U.S. Trend-Following Equity | 1 | 0.0095 |

**Replication basket** – VOO (cor=0.98) basket lead at lag 1: -0.019

| ticker | rep_wt | name                 | category              | etf_signal |
|--------|--------|----------------------|-----------------------|------------|
| VOO    | 1.145  | Vanguard S&P 500 ETF | U.S. Large Cap Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 7  | SPY    | 0.0125 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity    | LONG       |
| 53 | IWV    | 0.0125 | iShares Russell 3000 Fund                  | U.S. Total Market Equity | LONG       |
| 54 | IYY    | 0.0125 | iShares Dow Jones U.S. ETF                 | U.S. Total Market Equity | LONG       |
| 57 | SPYX   | 0.0125 | State Street SPDR S&P 500 Fossil Fuel R... | U.S. Equity              | LONG       |
| 59 | ESGU   | 0.0125 | iShares ESG Aware MSCI USA ETF             | U.S. ESG Equity          | LONG       |
| 60 | IWB    | 0.0125 | iShares Russell 1000 ETF                   | U.S. Large Cap Equity    | LONG       |
| 61 | IWL    | 0.0125 | iShares Russell Top 200 ETF                | U.S. Equity              | LONG       |
| 68 | VONE   | 0.0125 | Vanguard Russell 1000 ETF                  | U.S. Large Cap Equity    | LONG       |
| 69 | MGC    | 0.0125 | Vanguard Mega Cap ETF                      | U.S. Mega Cap Equity     | LONG       |
| 70 | VV     | 0.0125 | Vanguard Large-Cap ETF                     | U.S. Large Cap Equity    | LONG       |
| 71 | ILCB   | 0.0125 | iShares Morningstar Large-Cap ETF          | U.S. Large Cap Equity    | LONG       |
| 72 | ITOT   | 0.0125 | iShares Core S&P Total U.S. Stock Marke... | U.S. Total Market Equity | LONG       |
| 73 | IVV    | 0.0125 | iShares Core S&P 500 ETF                   | U.S. Large Cap Equity    | LONG       |
| 74 | SCHB   | 0.0125 | Schwab U.S. Broad Market ETF               | U.S. Total Market Equity | LONG       |
| 75 | SCHX   | 0.0125 | Schwab U.S. Large-Cap ETF                  | U.S. Large Cap Equity    | LONG       |



|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 76 | SPTM | 0.0125 | State Street SPDR Portfolio S&P 1500 Co... | U.S. Total Market Equity    | LONG |
| 77 | VOO  | 0.0125 | Vanguard S&P 500 ETF                       | U.S. Large Cap Equity       | LONG |
| 78 | VTI  | 0.0125 | Vanguard Total Stock Market ETF            | U.S. Total Market Equity    | LONG |
| 8  | QQQ  | 0.0124 | Invesco QQQ Trust, Series 1                | U.S. Large Cap Growth       | LONG |
| 25 | ONEQ | 0.0124 | Fidelity Nasdaq Composite Index ETF        | U.S. Equity                 | LONG |
| 27 | IWF  | 0.0124 | iShares Russell 1000 Growth Fund           | U.S. Large Cap Growth       | LONG |
| 36 | VONG | 0.0124 | Vanguard Russell 1000 Growth ETF           | U.S. Large Cap Growth       | LONG |
| 38 | ILCG | 0.0124 | iShares Morningstar Large-Cap Growth ETF   | U.S. Large Cap Growth       | LONG |
| 39 | IUSG | 0.0124 | iShares Core S&P U.S. Growth ETF           | U.S. Large Cap Growth       | LONG |
| 51 | DSI  | 0.0124 | iShares ESG MSCI KLD 400 ETF               | Large Cap Blend Equities    | LONG |
| 52 | SUSA | 0.0124 | iShares ESG Optimized MSCI USA ETF         | U.S. ESG Equity             | LONG |
| 56 | OEF  | 0.0124 | iShares S&P 100 Fund                       | U.S. Mega Cap Equity        | LONG |
| 65 | VTHR | 0.0124 | Vanguard Russell 3000 ETF                  | U.S. Total Market Equity    | LONG |
| 66 | GSLC | 0.0124 | Goldman Sachs ActiveBeta U.S. Large Cap... | Commodity                   | LONG |
| 26 | IWY  | 0.0123 | iShares Russell Top 200 Growth ETF         | Growth Equity               | LONG |
| 28 | IVW  | 0.0123 | iShares S&P 500 Growth ETF                 | U.S. Large Cap Growth       | LONG |
| 31 | VOOG | 0.0123 | Vanguard S&P 500 Growth ETF                | U.S. Large Cap Growth       | LONG |
| 37 | MGK  | 0.0123 | Vanguard Mega Cap Growth ETF               | U.S. Mega Cap Growth        | LONG |
| 40 | SCHG | 0.0123 | Schwab U.S. Large-Cap Growth ETF           | U.S. Large Cap Growth       | LONG |
| 41 | SPYG | 0.0123 | State Street SPDR Portfolio S&P 500 Gro... | U.S. Large Cap Growth       | LONG |
| 42 | VUG  | 0.0123 | Vanguard Growth ETF                        | U.S. Large Cap Growth       | LONG |
| 55 | LRGF | 0.0123 | iShares U.S. Equity Factor ETF             | U.S. Multifactor Equity     | LONG |
| 58 | XLG  | 0.0123 | Invesco S&P 500 Top 50 ETF                 | U.S. Mega Cap Equity        | LONG |
| 62 | QUAL | 0.0123 | iShares MSCI USA Quality Factor ETF        | U.S. Quality / Factor Eq... | LONG |
| 20 | IGM  | 0.0122 | iShares Expanded Tech Sector ETF           | Technology                  | LONG |
| 24 | NULG | 0.0122 | Nuveen ESG Large-Cap Growth ETF            | U.S. Large Cap Growth       | LONG |
| 44 | TILT | 0.0122 | FlexShares Mornigstar US Market Factors... | U.S. Multifactor Equity     | LONG |
| 49 | JHML | 0.0122 | John Hancock Multifactor Large Cap ETF     | U.S. Large Cap Equity       | LONG |
| 67 | EPS  | 0.0122 | WisdomTree U.S. LargeCap Fund              | Large Cap Value Equities    | LONG |
| 23 | IYW  | 0.0121 | iShares U.S. Technology ETF                | Technology                  | LONG |
| 30 | VGX  | 0.0121 | Vanguard Information Tech ETF              | Technology                  | LONG |
| 35 | FTEC | 0.0121 | Fidelity MSCI Information Technology In... | Technology                  | LONG |
| 46 | IOO  | 0.0121 | iShares Global 100 ETF                     | Global Equity               | LONG |
| 1  | XLK  | 0.0120 | State Street Technology Select Sector S... | Technology                  | LONG |
| 10 | QQEW | 0.0120 | First Trust Nasdaq-100 Select Equal Wei... | U.S. Equal Weight Equity    | LONG |
| 19 | PWB  | 0.0120 | Invesco Large Cap Growth ETF               | Growth Equity               | LONG |
| 21 | IXN  | 0.0120 | iShares Global Tech ETF                    | Technology                  | LONG |
| 48 | CATH | 0.0120 | Global X S&P 500 Catholic Values ETF       | Global Equity               | LONG |
| 16 | XNTK | 0.0119 | State Street SPDR NYSE Technology ETF      | Technology                  | LONG |
| 47 | QDF  | 0.0119 | FlexShares Quality Dividend Index Fund     | Dividend Equity             | LONG |
| 6  | AIQ  | 0.0118 | Global X Artificial Intelligence & Tech... | Technology                  | LONG |
| 11 | QTEC | 0.0118 | First Trust NASDAQ-100-Technology Secto... | Technology                  | LONG |
| 15 | FXL  | 0.0118 | First Trust Technology AlphaDEX            | Technology                  | LONG |
| 63 | QUS  | 0.0118 | State Street SPDR MSCI USA StrategicFac... | U.S. Quality / Factor Eq... | LONG |
| 64 | SPHQ | 0.0118 | Invesco S&P 500 Quality ETF                | U.S. Quality / Factor Eq... | LONG |

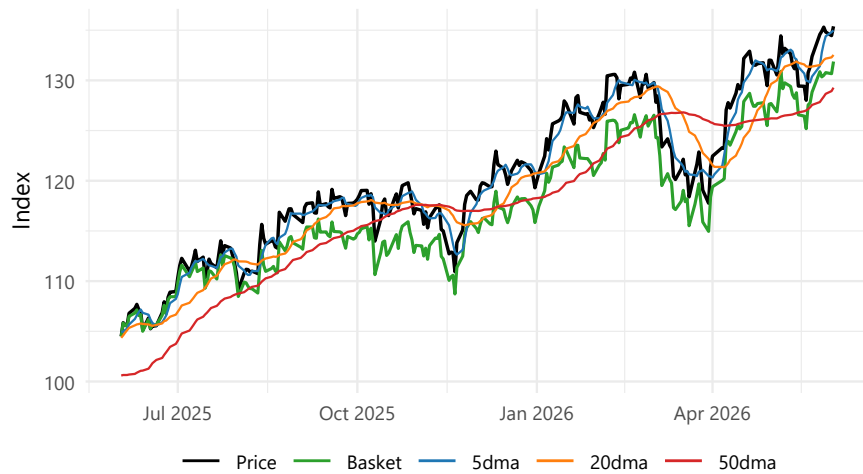
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 13 | TDIV | 0.0117 | First Trust NASDAQ Technology Dividend ... | Technology                  | LONG |
| 50 | DGRW | 0.0116 | WisdomTree U.S. Quality Dividend Growth... | U.S. Dividend Equity        | LONG |
| 22 | IYC  | 0.0115 | iShares U.S. Consumer Discretionary ETF    | Consumer Discretionary      | LONG |
| 29 | VCR  | 0.0114 | Vanguard Consumer Discretion ETF           | Consumer Discretionary      | FLAT |
| 34 | FDIS | 0.0114 | Fidelity MSCI Consumer Discretionary In... | Consumer Discretionary      | FLAT |
| 82 | OMFL | 0.0113 | Invesco Russell 1000 Dynamic Multifacto... | U.S. Multifactor Equity     | LONG |
| 2  | XLY  | 0.0112 | State Street Consumer Discretionary Sel... | Consumer Discretionary      | FLAT |
| 18 | QYLD | 0.0112 | Global X NASDAQ 100 Covered Call ETF       | Options Income              | FLAT |
| 9  | NXTG | 0.0111 | First Trust Indxx NextG ETF                | Technology                  | LONG |
| 43 | MOAT | 0.0111 | VanEck Morningstar Wide Moat ETF           | U.S. Moat Equity            | FLAT |
| 84 | QTUM | 0.0111 | Defiance Quantum ETF                       | Technology                  | LONG |
| 4  | SOXX | 0.0110 | iShares PHLX SOX Semiconductor Sector I... | Technology                  | LONG |
| 5  | SMH  | 0.0110 | VanEck Semiconductor ETF                   | Technology                  | LONG |
| 32 | VOX  | 0.0110 | Vanguard Communication Services ETF        | Communication Services      | LONG |
| 33 | FCOM | 0.0110 | Fidelity MSCI Communication Services In... | Communication Services      | LONG |
| 14 | XSD  | 0.0108 | State Street SPDR S&P Semiconductor ETF    | Technology                  | LONG |
| 79 | PSCT | 0.0108 | Invesco S&P SmallCap Information Techno... | U.S. Small Cap Technology   | LONG |
| 3  | XLC  | 0.0107 | State Street Communication Services Sel... | Communication Services      | FLAT |
| 12 | PSI  | 0.0107 | Invesco Semiconductors ETF                 | Technology                  | LONG |
| 81 | FTXL | 0.0107 | First Trust Nasdaq Semiconductor ETF       | Technology                  | LONG |
| 17 | PTNQ | 0.0106 | Pacer Trendpilot 100 ETF                   | Trend-Following Equity      | LONG |
| 83 | PTF  | 0.0106 | Invesco Dorsey Wright Technology Moment... | Technology                  | LONG |
| 80 | RTH  | 0.0102 | VanEck Retail ETF                          | Consumer Discretionary      | FLAT |
| 45 | PTLC | 0.0095 | Pacer Trendpilot US Large Cap ETF          | U.S. Trend-Following Equ... | LONG |

## Cluster 2 – U.S. Mid Cap Equity

**Members:** 79   **Var explained:** 0.879   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +42.7%)
- Latest sell\_cross: 2026-05-18
- Current 252d return: +29.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-10)



### Category breakdown

| category                              | n_members | abs_weight |
|---------------------------------------|-----------|------------|
| Industrials                           | 15        | 0.1771     |
| U.S. Small Cap Equity                 | 10        | 0.1323     |
| U.S. Mid Cap Equity                   | 6         | 0.0799     |
| U.S. Small Cap Value                  | 6         | 0.0786     |
| Consumer Discretionary                | 6         | 0.0696     |
| U.S. Mid Cap Value                    | 3         | 0.0397     |
| U.S. Small Cap Growth                 | 3         | 0.0397     |
| U.S. Mid Cap Growth                   | 3         | 0.0391     |
| Leveraged / Inverse                   | 2         | 0.0253     |
| U.S. Small Cap Momentum Equity        | 2         | 0.0249     |
| U.S. Momentum Equity                  | 2         | 0.0230     |
| Unknown                               | 1         | 0.0134     |
| U.S. Small Cap Fundamental Equity     | 1         | 0.0134     |
| U.S. Small Cap ESG Equity             | 1         | 0.0134     |
| U.S. Mid/Small Cap Equity             | 1         | 0.0134     |
| U.S. Small/Mid Cap Fundamental Equity | 1         | 0.0133     |
| U.S. Small Cap Multifactor Equity     | 1         | 0.0133     |
| Mid Cap Value Equities                | 1         | 0.0133     |
| Dividend Equity                       | 1         | 0.0131     |
| Small Cap Value Equities              | 1         | 0.0130     |
| U.S. Small/Mid Cap Equity             | 1         | 0.0130     |
| U.S. ESG Equity                       | 1         | 0.0130     |

|                                       |   |        |
|---------------------------------------|---|--------|
| U.S. Mid Cap Dividend Equity          | 1 | 0.0130 |
| U.S. Mid Cap Quality Equity           | 1 | 0.0129 |
| U.S. Equity                           | 1 | 0.0128 |
| Value Equity                          | 1 | 0.0127 |
| U.S. Small Cap Dividend Equity        | 1 | 0.0127 |
| U.S. Small Cap Industrials            | 1 | 0.0127 |
| U.S. Micro Cap Equity                 | 1 | 0.0123 |
| U.S. Mid Cap Momentum Equity          | 1 | 0.0123 |
| U.S. Small Cap Consumer Discretionary | 1 | 0.0120 |
| Momentum Equity                       | 1 | 0.0118 |

**Replication basket** – IJH (cor=0.99) basket lead at lag 1: -0.005

| ticker | rep_wt | name                         | category            | etf_signal |
|--------|--------|------------------------------|---------------------|------------|
| IJH    | 1.02   | iShares Core S&P Mid-Cap ETF | U.S. Mid Cap Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 13 | FNX    | 0.0134 | First Trust Mid Cap Core AlphaDEX Fund     | U.S. Mid/Small Cap Equity   | FLAT       |
| 17 | NUSC   | 0.0134 | Nuveen ESG Small-Cap ETF                   | U.S. Small Cap ESG Equity   | FLAT       |
| 20 | MDY    | 0.0134 | State Street SPDR S&P MIDCAP 400 ETF Tr... | U.S. Mid Cap Equity         | FLAT       |
| 27 | IVOO   | 0.0134 | Vanguard S&P Mid-Cap 400 ETF               |                             | FLAT       |
| 31 | IJH    | 0.0134 | iShares Core S&P Mid-Cap ETF               | U.S. Mid Cap Equity         | FLAT       |
| 32 | SPMD   | 0.0134 | State Street SPDR Portfolio S&P 400 Mid... | U.S. Mid Cap Equity         | FLAT       |
| 34 | VB     | 0.0134 | Vanguard Small-Cap ETF                     | U.S. Small Cap Equity       | FLAT       |
| 52 | FNDA   | 0.0134 | Schwab Fundamental U.S. Small Company E... | U.S. Small Cap Fundament... | FLAT       |
| 61 | VBR    | 0.0134 | Vanguard Small-Cap Value ETF               | U.S. Small Cap Value        | FLAT       |
| 12 | SCHM   | 0.0133 | Schwab U.S. Mid Cap ETF                    | U.S. Mid Cap Equity         | LONG       |
| 18 | PRFZ   | 0.0133 | Invesco RAFI US 1500 Small-Mid ETF         | U.S. Small/Mid Cap Funda... | FLAT       |
| 19 | SMLF   | 0.0133 | iShares U.S. Small-Cap Equity Factor ETF   | U.S. Small Cap Multifact... | FLAT       |
| 28 | VI00   | 0.0133 | Vanguard S&P Small-Cap 600 ETF             | U.S. Small Cap Equity       | FLAT       |
| 30 | IJR    | 0.0133 | iShares Core S&P Small-Cap ETF             | U.S. Small Cap Equity       | FLAT       |
| 33 | SPSM   | 0.0133 | State Street SPDR Portfolio S&P 600 Sma... | U.S. Small Cap Equity       | FLAT       |
| 35 | SCHA   | 0.0133 | Schwab U.S. Small-Cap ETF                  | U.S. Small Cap Equity       | LONG       |
| 47 | EZM    | 0.0133 | WisdomTree U.S. MidCap Fund                | Mid Cap Value Equities      | FLAT       |
| 4  | IWM    | 0.0132 | iShares Russell 2000 Index Fund            | U.S. Small Cap Equity       | FLAT       |
| 9  | JHMM   | 0.0132 | John Hancock Multifactor Mid Cap ETF       | U.S. Mid Cap Equity         | FLAT       |
| 14 | FYX    | 0.0132 | First Trust Small Cap Core AlphaDEX Fund   | U.S. Small Cap Equity       | FLAT       |
| 21 | IJT    | 0.0132 | iShares S&P SmallCap 600 Growth ETF        | U.S. Small Cap Growth       | FLAT       |
| 23 | VI0G   | 0.0132 | Vanguard S&P Small-Cap 600 Growth ETF      | U.S. Small Cap Growth       | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 26 | SLYG | 0.0132 | State Street SPDR S&P 600 Small Cap Gro... | U.S. Small Cap Growth       | FLAT |
| 29 | VTWO | 0.0132 | Vanguard Russell 2000 ETF                  | U.S. Small Cap Equity       | FLAT |
| 54 | IJJ  | 0.0132 | iShares S&P Mid-Cap 400 Value ETF          | U.S. Mid Cap Value          | FLAT |
| 56 | IVOV | 0.0132 | Vanguard S&P Mid-Cap 400 Value ETF         | U.S. Mid Cap Value          | FLAT |
| 57 | MDYV | 0.0132 | State Street SPDR S&P 400 Mid Cap Value... | U.S. Mid Cap Value          | FLAT |
| 71 | RWK  | 0.0132 | Invesco S&P MidCap 400 Revenue ETF         | U.S. Mid Cap Equity         | FLAT |
| 48 | PSC  | 0.0131 | Principal U.S. Small-Cap ETF               | U.S. Small Cap Equity       | FLAT |
| 53 | IWN  | 0.0131 | iShares Russell 2000 Value ETF             | U.S. Small Cap Value        | FLAT |
| 60 | VTWV | 0.0131 | Vanguard Russell 2000 Value ETF            | U.S. Small Cap Value        | FLAT |
| 78 | SDVY | 0.0131 | First Trust SMID Cap Rising Dividend Ac... | Dividend Equity             | FLAT |
| 7  | FXR  | 0.0130 | First Trust Industrials AlphaDEX           | Industrials                 | LONG |
| 22 | IJK  | 0.0130 | iShares S&P Mid-Cap 400 Growth ETF         | U.S. Mid Cap Growth         | LONG |
| 24 | IVOG | 0.0130 | Vanguard S&P Mid-Cap 400 Growth ETF        | U.S. Mid Cap Growth         | LONG |
| 25 | MDYG | 0.0130 | State Street SPDR S&P 400 Mid Cap Growt... | U.S. Mid Cap Growth         | LONG |
| 45 | DON  | 0.0130 | WisdomTree U.S. MidCap Dividend Fund       | U.S. Mid Cap Dividend Eq... | FLAT |
| 46 | EES  | 0.0130 | WisdomTree U.S. SmallCap Fund              | Small Cap Value Equities    | FLAT |
| 55 | IJS  | 0.0130 | iShares S&P SmallCap 600 Value ETF         | U.S. Small Cap Value        | LONG |
| 58 | SLYV | 0.0130 | State Street SPDR S&P 600 Small Cap Val... | U.S. Small Cap Value        | FLAT |
| 59 | VIOV | 0.0130 | Vanguard S&P Small-Cap 600 Value ETF       | U.S. Small Cap Value        | FLAT |
| 63 | ETHO | 0.0130 | Amplify Etho Climate Leadership U.S. ETF   | U.S. ESG Equity             | FLAT |
| 64 | RWJ  | 0.0130 | Invesco S&P SmallCap 600 Revenue ETF       | U.S. Small Cap Equity       | FLAT |
| 79 | ISMD | 0.0130 | Inspire Small/Mid Cap ETF                  | U.S. Small/Mid Cap Equity   | LONG |
| 67 | XMHQ | 0.0129 | Invesco S&P MidCap Quality ETF             | U.S. Mid Cap Quality Equ... | FLAT |
| 70 | RFV  | 0.0128 | Invesco S&P MidCap 400 Pure Value ETF      | Leveraged / Inverse         | FLAT |
| 72 | OMFS | 0.0128 | Invesco Russell 2000 Dynamic Multifacto... | U.S. Equity                 | FLAT |
| 10 | VIS  | 0.0127 | Vanguard Industrials ETF                   | Industrials                 | FLAT |
| 11 | FIDU | 0.0127 | Fidelity MSCI Industrials Index ETF        | Industrials                 | FLAT |
| 44 | DES  | 0.0127 | WisdomTree U.S. SmallCap Dividend Fund     | U.S. Small Cap Dividend ... | FLAT |
| 65 | PSCI | 0.0127 | Invesco S&P SmallCap Industrials ETF       | U.S. Small Cap Industria... | FLAT |
| 76 | XSMO | 0.0127 | Invesco S&P SmallCap Momentum ETF          | U.S. Small Cap Momentum ... | FLAT |
| 77 | FYT  | 0.0127 | First Trust Small Cap Value AlphaDEX Fu... | Value Equity                | FLAT |
| 8  | IYJ  | 0.0126 | iShares U.S. Industrials ETF               | Industrials                 | FLAT |
| 36 | FXD  | 0.0126 | First Trust Cons. Discret. AlphaDEX        | Consumer Discretionary      | LONG |
| 66 | PAVE | 0.0126 | Global X U.S. Infrastructure Developmen... | Industrials                 | FLAT |
| 73 | RZV  | 0.0125 | Invesco S&P SmallCap 600 Pure Value ETF    | Leveraged / Inverse         | FLAT |
| 1  | XLI  | 0.0123 | State Street Industrial Select Sector S... | Industrials                 | FLAT |
| 6  | XMMO | 0.0123 | Invesco S&P MidCap Momentum ETF            | U.S. Mid Cap Momentum Eq... | LONG |
| 16 | IWC  | 0.0123 | iShares Microcap ETF                       | U.S. Micro Cap Equity       | FLAT |
| 43 | RSPD | 0.0123 | Invesco S&P 500 Equal Weight Consumer D... | Consumer Discretionary      | LONG |
| 15 | DWAS | 0.0122 | Invesco Dorsey Wright SmallCap Momentum... | U.S. Small Cap Momentum ... | FLAT |
| 38 | FTXR | 0.0122 | First Trust Nasdaq Transportation ETF      | Industrials                 | LONG |
| 68 | VFMO | 0.0122 | Vanguard U.S. Momentum Factor ETF          | U.S. Momentum Equity        | LONG |
| 75 | PSCD | 0.0120 | Invesco S&P SmallCap Consumer Discretio... | U.S. Small Cap Consumer ... | LONG |
| 51 | XTN  | 0.0119 | State Street SPDR S&P Transportation ETF   | Industrials                 | FLAT |
| 69 | PKB  | 0.0119 | Invesco Building & Construction ETF        | Industrials                 | LONG |

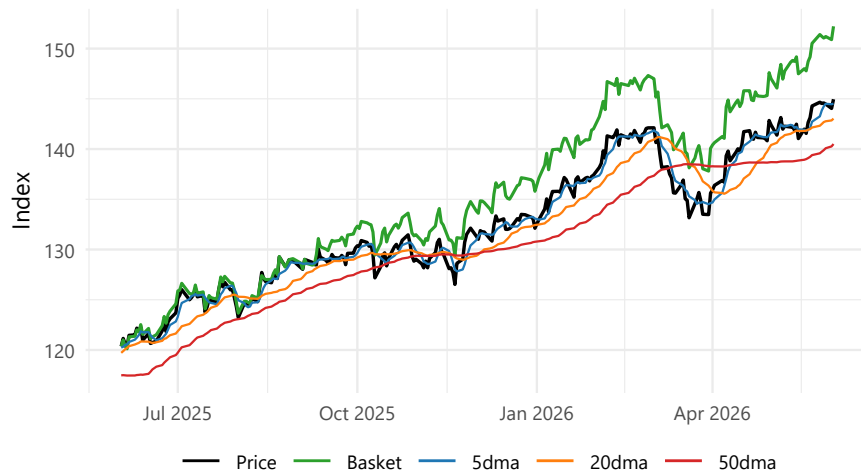
|    |      |        |                                            |                        |      |
|----|------|--------|--------------------------------------------|------------------------|------|
| 5  | FV   | 0.0118 | First Trust Dorsey Wright Focus 5 ETF      | Momentum Equity        | LONG |
| 2  | XRT  | 0.0116 | State Street SPDR S&P Retail ETF           | Consumer Discretionary | LONG |
| 42 | IYT  | 0.0116 | iShares U.S. Transportation ETF            | Industrials            | FLAT |
| 37 | PEJ  | 0.0115 | Invesco Leisure and Entertainment ETF      | Consumer Discretionary | LONG |
| 74 | PRN  | 0.0115 | Invesco Dorsey Wright Industrials Momen... | Industrials            | FLAT |
| 50 | XHB  | 0.0113 | State Street SPDR S&P Homebuilders ETF     | Consumer Discretionary | LONG |
| 49 | XAR  | 0.0108 | State Street SPDR S&P Aerospace & Defen... | Industrials            | LONG |
| 62 | QMOM | 0.0108 | Alpha Architect U.S. Quantitative Momen... | U.S. Momentum Equity   | FLAT |
| 39 | PPA  | 0.0107 | Invesco Aerospace & Defense ETF            | Industrials            | LONG |
| 3  | JETS | 0.0105 | U.S. Global Jets ETF                       | Industrials            | FLAT |
| 41 | ITB  | 0.0103 | iShares U.S. Home Construction ETF         | Consumer Discretionary | LONG |
| 40 | ITA  | 0.0100 | iShares U.S. Aerospace & Defense ETF       | Industrials            | LONG |

### Cluster 3 – U.S. Large Cap Value

**Members:** 77 **Var explained:** 0.878 **Signal:** LONG since 2026-04-13 (35 days, 3.8%)

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +22.4%)
- Latest sell\_cross: 2026-03-06
- Current 252d return: +20.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| U.S. Dividend Equity          | 22        | 0.2847     |
| U.S. Large Cap Value          | 10        | 0.1364     |
| Value Equity                  | 4         | 0.0527     |
| U.S. Low Volatility Equity    | 4         | 0.0490     |
| U.S. Multifactor Equity       | 3         | 0.0400     |
| U.S. Value Factor Equity      | 3         | 0.0394     |
| U.S. Mid Cap Equity           | 3         | 0.0394     |
| Dividend Equity               | 3         | 0.0380     |
| Utilities                     | 3         | 0.0364     |
| U.S. Fundamental Equity       | 2         | 0.0272     |
| U.S. Mid Cap Value            | 2         | 0.0271     |
| U.S. Equity                   | 2         | 0.0270     |
| U.S. Equal Weight Equity      | 2         | 0.0270     |
| U.S. Small Cap Low Volatility | 2         | 0.0247     |
| U.S. Mega Cap Value           | 1         | 0.0135     |
| U.S. Revenue Weighted Equity  | 1         | 0.0134     |
| Large Cap Blend Equities      | 1         | 0.0133     |
| U.S. Size Factor Equity       | 1         | 0.0132     |
| U.S. Buyback Equity           | 1         | 0.0131     |
| U.S. Large Cap Equity         | 1         | 0.0130     |
| U.S. Quality / Factor Equity  | 1         | 0.0128     |
| U.S. Mid Cap Low Volatility   | 1         | 0.0128     |

|                        |   |        |
|------------------------|---|--------|
| U.S. Defensive Equity  | 1 | 0.0125 |
| Global Equity          | 1 | 0.0120 |
| Communication Services | 1 | 0.0110 |
| Real Estate            | 1 | 0.0105 |

**Replication basket** – VTV (cor=0.98) basket lead at lag 1: 0.006

| ticker | rep_wt | name               | category             | etf_signal |
|--------|--------|--------------------|----------------------|------------|
| VTV    | 1.008  | Vanguard Value ETF | U.S. Large Cap Value | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 29 | IWD    | 0.0137 | iShares Russell 1000 Value ETF             | U.S. Large Cap Value        | LONG       |
| 32 | JPUS   | 0.0137 | JPMorgan Diversified Return U.S. Equity... | U.S. Multifactor Equity     | FLAT       |
| 37 | VONV   | 0.0137 | Vanguard Russell 1000 Value ETF            | U.S. Large Cap Value        | LONG       |
| 44 | SCHV   | 0.0137 | Schwab U.S. Large-Cap Value ETF            | U.S. Large Cap Value        | LONG       |
| 3  | RSP    | 0.0136 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity    | LONG       |
| 14 | PRF    | 0.0136 | Invesco RAFI US 1000 ETF                   | U.S. Fundamental Equity     | LONG       |
| 19 | CFA    | 0.0136 | VictoryShares US 500 Volatility Wtd ETF    | U.S. Multifactor Equity     | FLAT       |
| 21 | NULV   | 0.0136 | Nuveen ESG Large-Cap Value ETF             | U.S. Large Cap Value        | LONG       |
| 25 | FNDX   | 0.0136 | Schwab Fundamental U.S. Large Company E... | U.S. Fundamental Equity     | LONG       |
| 31 | IVE    | 0.0136 | iShares S&P 500 Value ETF                  | U.S. Large Cap Value        | LONG       |
| 35 | VOOV   | 0.0136 | Vanguard S&P 500 Value ETF                 | U.S. Large Cap Value        | LONG       |
| 41 | ILCV   | 0.0136 | iShares Morningstar Large-Cap Value ETF    | U.S. Large Cap Value        | LONG       |
| 43 | IUSV   | 0.0136 | iShares Core S&P U.S. Value ETF            | U.S. Large Cap Value        | LONG       |
| 45 | SPYV   | 0.0136 | State Street SPDR Portfolio S&P 500 Val... | U.S. Large Cap Value        | LONG       |
| 47 | VTV    | 0.0136 | Vanguard Value ETF                         | U.S. Large Cap Value        | LONG       |
| 63 | VOE    | 0.0136 | Vanguard Mid-Cap Value ETF                 | U.S. Mid Cap Value          | FLAT       |
| 68 | DTD    | 0.0136 | WisdomTree U.S. Total Dividend Fund        | U.S. Dividend Equity        | LONG       |
| 73 | DGRO   | 0.0136 | iShares Core Dividend Growth ETF           | U.S. Dividend Equity        | LONG       |
| 17 | AIVL   | 0.0135 | WisdomTree U.S. AI Enhanced Value Fund     | U.S. Value Factor Equity    | LONG       |
| 27 | IWX    | 0.0135 | iShares Russell Top 200 Value ETF          | Value Equity                | LONG       |
| 28 | ONEV   | 0.0135 | State Street SPDR Russell 1000 Low Vola... | U.S. Equity                 | FLAT       |
| 38 | MGV    | 0.0135 | Vanguard Mega Cap Value ETF                | U.S. Mega Cap Value         | LONG       |
| 40 | VYM    | 0.0135 | Vanguard High Dividend Yield ETF           | U.S. Dividend Equity        | LONG       |
| 59 | IWS    | 0.0135 | iShares Russell Mid-Cap Value ETF          | U.S. Mid Cap Value          | FLAT       |
| 67 | DLN    | 0.0135 | WisdomTree U.S. LargeCap Dividend Fund     | U.S. Dividend Equity        | LONG       |
| 76 | JPME   | 0.0135 | JPMorgan Diversified Return U.S. Mid Ca... | U.S. Equity                 | FLAT       |
| 15 | RWL    | 0.0134 | Invesco S&P 500 Revenue ETF                | U.S. Revenue Weighted Eq... | LONG       |
| 48 | FTA    | 0.0134 | First Trust Large Cap Value AlphaDEX Fu... | Value Equity                | LONG       |



|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 60 | ONEY | 0.0134 | State Street SPDR Russell 1000 Yield Fo... | U.S. Dividend Equity        | LONG |
| 65 | PFM  | 0.0134 | Invesco Dividend Achievers ETF             | U.S. Dividend Equity        | LONG |
| 6  | FEX  | 0.0133 | First Trust Large Cap Core AlphaDEX Fund   | Large Cap Blend Equities    | LONG |
| 26 | EQAL | 0.0133 | Invesco Russell 1000 Equal Weight ETF      | U.S. Equal Weight Equity    | FLAT |
| 4  | FVD  | 0.0132 | First Trust VL Dividend                    | U.S. Dividend Equity        | LONG |
| 22 | SDY  | 0.0132 | State Street SPDR S&P Dividend ETF         | U.S. Dividend Equity        | LONG |
| 23 | FDVV | 0.0132 | Fidelity High Dividend ETF                 | U.S. Dividend Equity        | LONG |
| 33 | SIZE | 0.0132 | iShares MSCI USA Size Factor ETF           | U.S. Size Factor Equity     | LONG |
| 39 | SCHD | 0.0132 | Schwab US Dividend Equity ETF              | U.S. Dividend Equity        | LONG |
| 42 | IMCB | 0.0132 | iShares Morningstar Mid-Cap ETF            | U.S. Mid Cap Equity         | LONG |
| 53 | DVY  | 0.0132 | iShares Select Dividend ETF                | U.S. Dividend Equity        | FLAT |
| 74 | VIG  | 0.0132 | Vanguard Div Appreciation ETF              | U.S. Dividend Equity        | LONG |
| 77 | JVAL | 0.0132 | JPMorgan U.S. Value Factor ETF             | U.S. Value Factor Equity    | LONG |
| 5  | PKW  | 0.0131 | Invesco BuyBack Achievers ETF              | U.S. Buyback Equity         | LONG |
| 30 | IWR  | 0.0131 | iShares Russell Mid-Cap ETF                | U.S. Mid Cap Equity         | LONG |
| 46 | VO   | 0.0131 | Vanguard Mid-Cap ETF                       | U.S. Mid Cap Equity         | LONG |
| 51 | SDOG | 0.0131 | ALPS Sector Dividend Dogs ETF              | U.S. Dividend Equity        | LONG |
| 10 | PWV  | 0.0130 | Invesco Large Cap Value ETF                | Value Equity                | LONG |
| 70 | DIA  | 0.0130 | State Street SPDR Dow Jones Industrial ... | U.S. Large Cap Equity       | LONG |
| 11 | RDVY | 0.0129 | First Trust Rising Dividend Achievers E... | U.S. Dividend Equity        | FLAT |
| 62 | SPYD | 0.0129 | State Street SPDR Portfolio S&P 500 Hig... | U.S. Dividend Equity        | LONG |
| 72 | LGLV | 0.0129 | State Street SPDR US Large Cap Low Vola... | U.S. Low Volatility Equi... | FLAT |
| 13 | REGL | 0.0128 | ProShares S&P MidCap 400 Dividend Arist... | Dividend Equity             | FLAT |
| 34 | VLUE | 0.0128 | iShares MSCI USA Value Factor ETF          | U.S. Value Factor Equity    | LONG |
| 57 | XMLV | 0.0128 | Invesco S&P MidCap Low Volatility ETF      | U.S. Mid Cap Low Volatil... | FLAT |
| 64 | FTCS | 0.0128 | First Trust Capital Strength ETF           | U.S. Quality / Factor Eq... | LONG |
| 66 | OUSA | 0.0128 | ALPS O'Shares U.S. Quality Dividend ETF    | Dividend Equity             | LONG |
| 16 | DHS  | 0.0127 | WisdomTree U.S. High Dividend Fund         | U.S. Dividend Equity        | LONG |
| 20 | CFO  | 0.0127 | VictoryShares US 500 Enhanced Volatilit... | U.S. Multifactor Equity     | FLAT |
| 55 | RPV  | 0.0127 | Invesco S&P 500 Pure Value ETF             | Value Equity                | LONG |
| 61 | SMMV | 0.0127 | iShares MSCI USA Small-Cap Min Vol Fact... | U.S. Small Cap Low Volat... | FLAT |
| 18 | CDC  | 0.0125 | VictoryShares US EQ Income Enhanced Vol... | U.S. Defensive Equity       | FLAT |
| 71 | USMV | 0.0125 | iShares MSCI USA Min Vol Factor ETF        | U.S. Low Volatility Equi... | LONG |
| 9  | FIW  | 0.0124 | First Trust Water ETF                      | Utilities                   | LONG |
| 54 | RDIV | 0.0124 | Invesco S&P Ultra Dividend Revenue ETF     | U.S. Dividend Equity        | LONG |
| 7  | PHO  | 0.0123 | Invesco Water Resources ETF                | Utilities                   | FLAT |
| 49 | PEY  | 0.0123 | Invesco High Yield Equity Dividend Achi... | U.S. Dividend Equity        | FLAT |
| 56 | SPHD | 0.0123 | Invesco S&P 500 High Dividend Low Volat... | Dividend Equity             | LONG |
| 12 | FDL  | 0.0122 | First Trust Morningstar ETF                | U.S. Dividend Equity        | LONG |
| 50 | DIV  | 0.0122 | Global X Super Dividend ETF                | U.S. Dividend Equity        | FLAT |
| 2  | ACWV | 0.0120 | iShares MSCI Global Min Vol Factor ETF     | Global Equity               | LONG |
| 24 | LVHD | 0.0120 | Franklin U.S. Low Volatility High Divid... | U.S. Low Volatility Equi... | LONG |
| 58 | XSLV | 0.0120 | Invesco S&P SmallCap Low Volatility ETF    | U.S. Small Cap Low Volat... | FLAT |
| 36 | HDV  | 0.0119 | iShares Core High Dividend ETF             | U.S. Dividend Equity        | LONG |
| 52 | SMDV | 0.0119 | ProShares Russell 2000 Dividend Growers... | U.S. Dividend Equity        | FLAT |

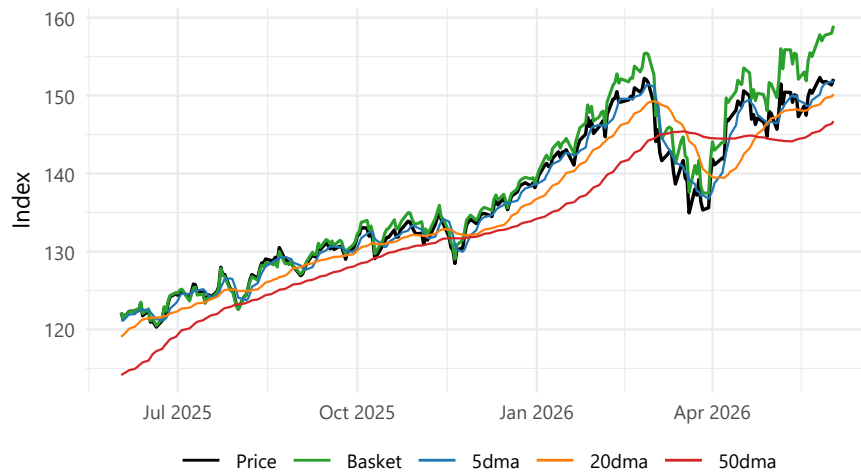
|    |      |        |                                     |                             |      |
|----|------|--------|-------------------------------------|-----------------------------|------|
| 8  | CGW  | 0.0117 | Invesco S&P Global Water Index ETF  | Utilities                   | FLAT |
| 69 | SPLV | 0.0116 | Invesco S&P 500 Low Volatility ETF  | U.S. Low Volatility Equi... | FLAT |
| 75 | IYZ  | 0.0110 | iShares U.S. Telecommunications ETF | Communication Services      | FLAT |
| 1  | REM  | 0.0105 | iShares Mortgage Real Estate ETF    | Real Estate                 | FLAT |

#### Cluster 4 – Developed ex-US Equity

**Members:** 61   **Var explained:** 0.898   **Signal:** LONG since 2026-04-10 (36 days, 3.4%)

##### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell\_cross: 2026-03-05
- Current 252d return: +25.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)



##### Category breakdown

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| International Equity   | 19        | 0.3159     |
| Developed ex-US Equity | 9         | 0.1528     |

|                               |   |        |
|-------------------------------|---|--------|
| Global ex-US Equity           | 5 | 0.0849 |
| Europe Equity                 | 4 | 0.0675 |
| International Dividend Equity | 4 | 0.0660 |
| Alternatives                  | 4 | 0.0614 |
| Eurozone Equity               | 3 | 0.0489 |
| Commodity                     | 1 | 0.0171 |
| Global ex-US Small Cap Equity | 1 | 0.0166 |
| Germany Equity                | 1 | 0.0162 |
| France Equity                 | 1 | 0.0162 |
| Italy Equity                  | 1 | 0.0159 |
| United Kingdom Equity         | 1 | 0.0158 |
| Europe Financials             | 1 | 0.0158 |
| Netherlands Equity            | 1 | 0.0157 |
| Sweden Equity                 | 1 | 0.0156 |
| Spain Equity                  | 1 | 0.0150 |
| Austria Equity                | 1 | 0.0147 |
| Belgium Equity                | 1 | 0.0146 |
| Poland Equity                 | 1 | 0.0135 |

**Replication basket** – VEA (cor=0.99) basket lead at lag 1: -0.012

| ticker | rep_wt | name                                | category               | etf_signal |
|--------|--------|-------------------------------------|------------------------|------------|
| VEA    | 0.971  | Vanguard FTSE Developed Markets ETF | Developed ex-US Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2  | EFA    | 0.0172 | iShares MSCI EAFE ETF                      | Developed ex-US Equity      | FLAT       |
| 3  | VEA    | 0.0172 | Vanguard FTSE Developed Markets ETF        | Developed ex-US Equity      | LONG       |
| 41 | ESGD   | 0.0172 | iShares ESG Aware MSCI EAFE ETF            | International Equity        | FLAT       |
| 51 | IEFA   | 0.0172 | iShares Core MSCI EAFE ETF                 | Developed ex-US Equity      | FLAT       |
| 52 | SCHF   | 0.0172 | Schwab International Equity ETF            | Developed ex-US Equity      | LONG       |
| 53 | SPDW   | 0.0172 | State Street SPDR Portfolio Developed W... | Developed ex-US Equity      | LONG       |
| 25 | TLTD   | 0.0171 | FlexShares Morningstar Developed Market... | International Equity        | FLAT       |
| 40 | GSIE   | 0.0171 | Goldman Sachs ActiveBeta International ... | Commodity                   | FLAT       |
| 6  | IEV    | 0.0170 | iShares Europe ETF                         | Europe Equity               | FLAT       |
| 16 | DWM    | 0.0170 | WisdomTree International Equity Fund       | International Dividend E... | FLAT       |
| 19 | PXF    | 0.0170 | Invesco RAFI Developed Markets ex-U.S. ... | International Equity        | LONG       |
| 29 | ACWX   | 0.0170 | iShares MSCI ACWI ex U.S. ETF              | Global ex-US Equity         | LONG       |
| 32 | INTF   | 0.0170 | iShares International Equity Factor ETF    | International Equity        | FLAT       |
| 46 | IEUR   | 0.0170 | iShares Core MSCI Europe ETF               | Europe Equity               | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 47 | IXUS | 0.0170 | iShares Core MSCI Total International S... | Global ex-US Equity         | LONG |
| 48 | VEU  | 0.0170 | Vanguard FTSE All World Ex US ETF          | Global ex-US Equity         | LONG |
| 49 | VGK  | 0.0170 | Vanguard FTSEEuropean ETF                  | Europe Equity               | FLAT |
| 50 | VXUS | 0.0170 | Vanguard Total International Stock ETF     | Global ex-US Equity         | LONG |
| 30 | CWI  | 0.0169 | State Street SPDR MSCI ACWI ex-US ETF      | Global ex-US Equity         | LONG |
| 33 | IQLT | 0.0169 | iShares MSCI Intl Quality Factor ETF       | International Equity        | FLAT |
| 35 | QEFA | 0.0169 | State Street SPDR MSCI EAFE StrategicFa... | International Equity        | FLAT |
| 39 | FNDF | 0.0169 | Schwab Fundamental International Equity... | International Equity        | LONG |
| 23 | EFV  | 0.0168 | iShares MSCI EAFE Value ETF                | Developed ex-US Equity      | FLAT |
| 14 | EZU  | 0.0167 | iShares MSCI Eurozone ETF                  | Eurozone Equity             | FLAT |
| 18 | IQDF | 0.0167 | FlexShares International Quality Divide... | International Equity        | LONG |
| 22 | SCZ  | 0.0167 | iShares MSCI EAFE Small-Cap ETF            | Developed ex-US Equity      | LONG |
| 24 | FNDC | 0.0167 | Schwab Fundamental International Small ... | International Equity        | LONG |
| 26 | JPIN | 0.0167 | JPMorgan Diversified Return Internation... | International Equity        | FLAT |
| 37 | RODM | 0.0167 | Hartford Multifactor Developed Markets ... | International Equity        | FLAT |
| 38 | VYMI | 0.0167 | Vanguard International High Dividend Yi... | International Dividend E... | LONG |
| 44 | SCHC | 0.0167 | Schwab International Small-Cap Equity E... | Developed ex-US Equity      | FLAT |
| 56 | HFXI | 0.0167 | NYLI FTSE International Equity Currency... | International Equity        | LONG |
| 20 | EFG  | 0.0166 | iShares MSCI EAFE Growth ETF               | Developed ex-US Equity      | FLAT |
| 34 | IVLU | 0.0166 | iShares MSCI Intl Value Factor ETF         | International Equity        | FLAT |
| 36 | FEZ  | 0.0166 | State Street SPDR EURO STOXX 50 ETF        | Eurozone Equity             | FLAT |
| 45 | VSS  | 0.0166 | Vanguard FTSE All-Wld ex-US SmCp Idx ETF   | Global ex-US Small Cap E... | FLAT |
| 4  | FEP  | 0.0165 | First Trust Europe AlphaDEX Fund           | Europe Equity               | LONG |
| 7  | DLS  | 0.0165 | WisdomTree International SmallCap Fund     | International Equity        | LONG |
| 31 | IMTM | 0.0164 | iShares MSCI Intl Momentum Factor ETF      | International Equity        | LONG |
| 43 | VIGI | 0.0163 | Vanguard International Dividend Appreci... | International Dividend E... | LONG |
| 1  | EWG  | 0.0162 | iShares MSCI Germany Index Fund            | Germany Equity              | LONG |
| 12 | EWQ  | 0.0162 | iShares MSCI France Index Fund             | France Equity               | FLAT |
| 21 | GWX  | 0.0162 | State Street SPDR S&P International Sma... | International Equity        | FLAT |
| 5  | VIDI | 0.0160 | Vident International Equity Strategy ETF   | International Equity        | LONG |
| 59 | IPKW | 0.0160 | Invesco International BuyBack Achievers... | International Equity        | LONG |
| 15 | IDV  | 0.0159 | iShares International Select Dividend E... | International Dividend E... | LONG |
| 61 | EWI  | 0.0159 | iShares MSCI Italy ETF                     | Italy Equity                | LONG |
| 13 | EWU  | 0.0158 | iShares MSCI United Kingdom ETF            | United Kingdom Equity       | FLAT |
| 17 | EUFN | 0.0158 | iShares MSCI Europe Financials ETF         | Europe Financials           | FLAT |
| 42 | HDEF | 0.0158 | Xtrackers MSCI EAFE High Dividend Yield... | International Equity        | FLAT |
| 27 | DBEF | 0.0157 | Xtrackers MSCI EAFE Hedged Equity ETF      | Alternatives                | FLAT |
| 58 | EWN  | 0.0157 | iShares MSCI Netherlands Index Fund        | Netherlands Equity          | LONG |
| 10 | HEZU | 0.0156 | iShares Currency Hedged MSCI Eurozone E... | Eurozone Equity             | FLAT |
| 28 | HEFA | 0.0156 | iShares Currency Hedged MSCI EAFE ETF      | Alternatives                | FLAT |
| 55 | EWD  | 0.0156 | iShares MSCI Sweden ETF                    | Sweden Equity               | LONG |
| 8  | HEDJ | 0.0151 | WisdomTree Europe Hedged Equity Fund       | Alternatives                | FLAT |
| 9  | IHDG | 0.0151 | WisdomTree International Hedged Quality... | Alternatives                | LONG |
| 11 | EWP  | 0.0150 | iShares MSCI Spain ETF                     | Spain Equity                | FLAT |
| 57 | EWO  | 0.0147 | iShares MSCI Austria ETF                   | Austria Equity              | LONG |

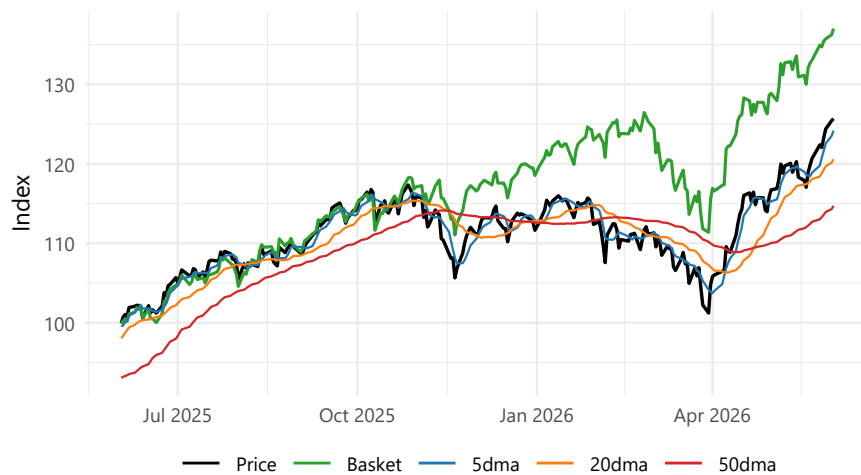
|    |      |        |                          |                |      |
|----|------|--------|--------------------------|----------------|------|
| 60 | EWK  | 0.0146 | iShares MSCI Belgium ETF | Belgium Equity | FLAT |
| 54 | EPOL | 0.0135 | iShares MSCI Poland ETF  | Poland Equity  | FLAT |

## Cluster 5 – Global Equity

**Members:** 49   **Var explained:** 0.808   **Signal:** LONG since 2026-04-13 (35 days, 13.2%)

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +25.7%)
- Latest sell\_cross: 2026-03-06
- Current 252d return: +26.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Technology          | 16        | 0.3189     |
| Global Equity       | 5         | 0.1086     |
| Asset Allocation    | 4         | 0.0807     |
| U.S. Mid Cap Growth | 3         | 0.0662     |

|                             |   |        |
|-----------------------------|---|--------|
| U.S. Small Cap Growth       | 3 | 0.0647 |
| Growth Equity               | 2 | 0.0428 |
| Convertible Bond            | 2 | 0.0415 |
| IPO Equity                  | 2 | 0.0415 |
| Financials                  | 2 | 0.0408 |
| Consumer Discretionary      | 2 | 0.0389 |
| U.S. Extended Market Equity | 1 | 0.0220 |
| U.S. Equity                 | 1 | 0.0214 |
| Momentum Equity             | 1 | 0.0207 |
| Alternatives                | 1 | 0.0199 |
| U.S. Momentum Equity        | 1 | 0.0195 |
| International IPO Equity    | 1 | 0.0186 |
| Israel Equity               | 1 | 0.0169 |
| Singapore Equity            | 1 | 0.0163 |

**Replication basket** – VT (cor=0.94) basket lead at lag 1: 0.002

| ticker | rep_wt | name                                 | category      | etf_signal |
|--------|--------|--------------------------------------|---------------|------------|
| VT     | 1.224  | Vanguard Total World Stock Index ETF | Global Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 37 | IWP    | 0.0221 | iShares Russell Midcap Growth ETF          | U.S. Mid Cap Growth         | LONG       |
| 40 | VOT    | 0.0221 | Vanguard Mid-Cap Growth ETF                | U.S. Mid Cap Growth         | LONG       |
| 41 | IMCG   | 0.0221 | iShares Morningstar Mid-Cap Growth ETF     | U.S. Mid Cap Growth         | LONG       |
| 44 | VBK    | 0.0220 | Vanguard Small-Cap Growth ETF              | U.S. Small Cap Growth       | LONG       |
| 46 | VXF    | 0.0220 | Vanguard Extended Market ETF               | U.S. Extended Market Equ... | FLAT       |
| 23 | XT     | 0.0219 | iShares Future Exponential Technologies... | Technology                  | LONG       |
| 21 | VT     | 0.0218 | Vanguard Total World Stock Index ETF       | Global Equity               | LONG       |
| 13 | ACWI   | 0.0217 | iShares MSCI ACWI ETF                      | Global Equity               | LONG       |
| 18 | URTH   | 0.0217 | iShares MSCI World ETF                     | Global Equity               | LONG       |
| 19 | CRBN   | 0.0217 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity               | LONG       |
| 20 | SPGM   | 0.0217 | State Street SPDR Portfolio MSCI Global... | Global Equity               | LONG       |
| 14 | AOA    | 0.0216 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation            | LONG       |
| 45 | ISCG   | 0.0216 | iShares Morningstar Small-Cap Growth ETF   | Growth Equity               | FLAT       |
| 42 | IWO    | 0.0214 | iShares Russell 2000 Growth Fund           | U.S. Small Cap Growth       | LONG       |
| 47 | SPHB   | 0.0214 | Invesco S&P 500 High Beta ETF              | U.S. Equity                 | LONG       |
| 4  | ROBO   | 0.0213 | ROBO Global Robotics and Automation Ind... | Technology                  | LONG       |
| 43 | VTWG   | 0.0213 | Vanguard Russell 2000 Growth ETF           | U.S. Small Cap Growth       | LONG       |
| 32 | FTC    | 0.0212 | First Trust Large Cap Growth AlphaDEX F... | Technology                  | LONG       |

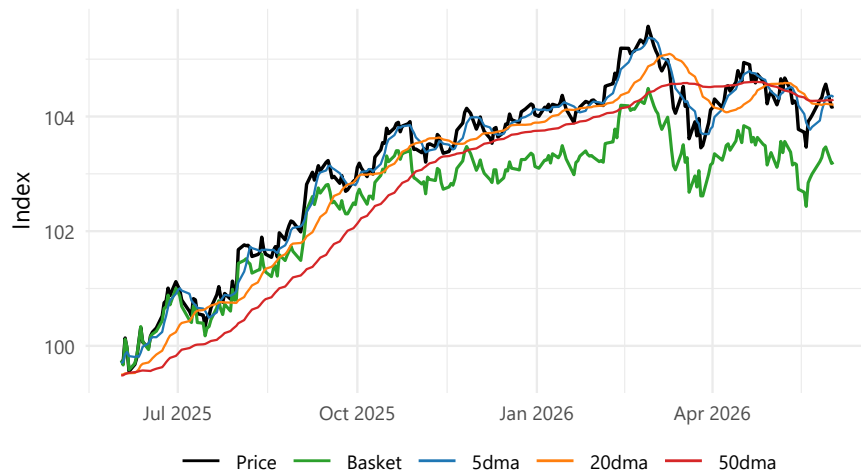
|    |      |        |                                            |                          |      |
|----|------|--------|--------------------------------------------|--------------------------|------|
| 17 | AOR  | 0.0211 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation         | LONG |
| 34 | FPX  | 0.0211 | First Trust US Equity Opportunities ETF    | IPO Equity               | LONG |
| 36 | RPG  | 0.0211 | Invesco S&P 500 Pure Growth ETF            | Growth Equity            | LONG |
| 35 | CWB  | 0.0210 | State Street SPDR Bloomberg Convertible... | Convertible Bond         | LONG |
| 3  | BOTZ | 0.0209 | Global X Robotics & Artificial Intellig... | Technology               | LONG |
| 31 | FINX | 0.0208 | Global X FinTech ETF                       | Financials               | FLAT |
| 33 | PDP  | 0.0207 | Invesco Dorsey Wright Momentum ETF         | Momentum Equity          | LONG |
| 9  | ARKQ | 0.0205 | ARK Autonomous Technology & Robotics ETF   | Technology               | LONG |
| 27 | PNQI | 0.0205 | Invesco Nasdaq Internet ETF                | Technology               | FLAT |
| 28 | FDN  | 0.0205 | First Trust DJ Internet Index Fund         | Technology               | LONG |
| 38 | ICVT | 0.0205 | iShares Convertible Bond ETF               | Convertible Bond         | LONG |
| 26 | IPO  | 0.0204 | Renaissance IPO ETF                        | IPO Equity               | LONG |
| 2  | SKYY | 0.0203 | First Trust Cloud Computing ETF            | Technology               | LONG |
| 24 | IBUY | 0.0202 | Amplify Online Retail ETF                  | Consumer Discretionary   | LONG |
| 8  | ARKW | 0.0201 | ARK Next Generation Internet ETF           | Technology               | LONG |
| 7  | ARKK | 0.0200 | ARK Innovation ETF                         | Technology               | FLAT |
| 29 | IPAY | 0.0200 | Amplify Digital Payments ETF               | Financials               | FLAT |
| 11 | QAI  | 0.0199 | NYLI Hedge Multi-Strategy Tracker ETF      | Alternatives             | LONG |
| 16 | AOM  | 0.0198 | iShares Core 40/60 Moderate Allocation ... | Asset Allocation         | LONG |
| 22 | PRNT | 0.0198 | 3D Printing (The) ETF                      | Technology               | LONG |
| 1  | IGV  | 0.0195 | iShares Expanded Tech-Software Sector E... | Technology               | LONG |
| 39 | MTUM | 0.0195 | iShares MSCI USA Momentum Factor ETF       | U.S. Momentum Equity     | LONG |
| 25 | CIBR | 0.0190 | First Trust NASDAQ Cybersecurity ETF       | Technology               | LONG |
| 6  | HACK | 0.0188 | Amplify Cybersecurity ETF                  | Technology               | LONG |
| 10 | ESPO | 0.0187 | VanEck Video Gaming and eSports ETF        | Consumer Discretionary   | FLAT |
| 30 | FPXI | 0.0186 | First Trust International Equity Opport... | International IPO Equity | LONG |
| 5  | BLOK | 0.0183 | Amplify Blockchain Technology ETF          | Technology               | LONG |
| 15 | AOK  | 0.0183 | iShares Core 30/70 Conservative Allocat... | Asset Allocation         | LONG |
| 48 | EIS  | 0.0169 | iShares MSCI Israel ETF                    | Israel Equity            | FLAT |
| 12 | EWS  | 0.0163 | iShares MSCI Singapore ETF                 | Singapore Equity         | LONG |
| 49 | BLCN | 0.0162 | Siren NexGen Economy ETF                   | Technology               | LONG |

## Cluster 6 – Aggregate Fixed Income

**Members:** 43   **Var explained:** 0.842   **Signal:** LONG since 2026-05-29 (2 days, -0.4%)

### Signal logic:

- Latest buy\_cross: 2026-05-29 – filter **PASS** (252d at date = +5.1%)
- Latest sell\_cross: 2026-05-13
- Current 252d return: +4.3% – filter **ALLows** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-13)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Fixed Income                    | 13        | 0.3081     |
| Aggregate Fixed Income          | 6         | 0.1478     |
| Investment Grade Corporate Bond | 6         | 0.1441     |
| Mortgage-Backed Fixed Income    | 6         | 0.1326     |
| Active Fixed Income             | 4         | 0.0942     |
| Cash / Ultra-Short Fixed Income | 3         | 0.0667     |
| Target Maturity Bond            | 2         | 0.0422     |
| International Fixed Income      | 2         | 0.0417     |
| Agency Fixed Income             | 1         | 0.0227     |

**Replication basket** – BND (cor=0.99) basket lead at lag 1: 0.011

| ticker | rep_wt | name                           | category               | etf_signal |
|--------|--------|--------------------------------|------------------------|------------|
| BND    | 0.729  | Vanguard Total Bond Market ETF | Aggregate Fixed Income | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)



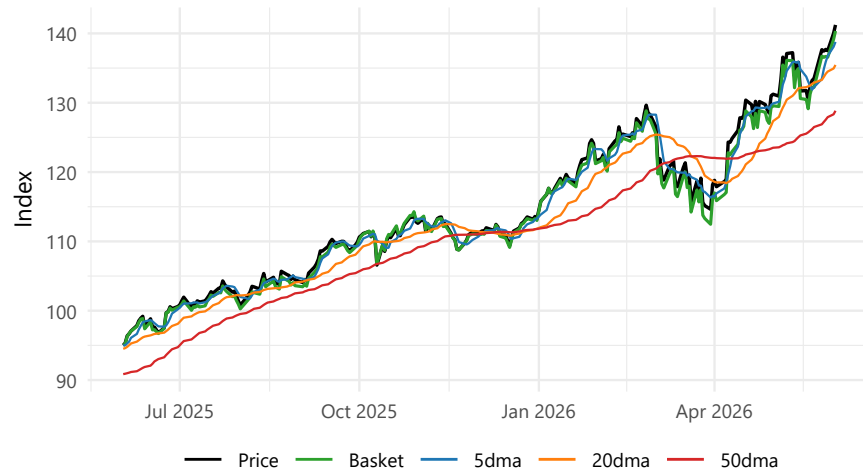
|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 13 | BIV    | 0.0250 | Vanguard Intermediate-Term Bond ETF        | Fixed Income                | FLAT       |
| 16 | AGG    | 0.0249 | iShares Core U.S. Aggregate Bond ETF       | Aggregate Fixed Income      | LONG       |
| 33 | VGIT   | 0.0249 | Vanguard Intermediate-Term Treasury ETF    | Fixed Income                | FLAT       |
| 17 | BND    | 0.0248 | Vanguard Total Bond Market ETF             | Aggregate Fixed Income      | LONG       |
| 19 | SCHZ   | 0.0248 | Schwab US Aggregate Bond ETF               | Aggregate Fixed Income      | LONG       |
| 29 | IEI    | 0.0248 | iShares 3-7 Year Treasury Bond ETF         | Fixed Income                | FLAT       |
| 30 | SPTI   | 0.0248 | State Street SPDR Portfolio Intermediat... | Fixed Income                | FLAT       |
| 32 | SCHR   | 0.0248 | Schwab Intermediate-Term U.S. Treasury ... | Fixed Income                | FLAT       |
| 12 | IUSB   | 0.0247 | iShares Core Universal USD Bond ETF        | Aggregate Fixed Income      | LONG       |
| 20 | SPAB   | 0.0247 | State Street SPDR Portfolio Aggregate B... | Aggregate Fixed Income      | LONG       |
| 1  | IEF    | 0.0246 | iShares 7-10 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 6  | GVI    | 0.0246 | iShares Intermediate Government/Credit ... | Fixed Income                | FLAT       |
| 11 | ISTB   | 0.0245 | iShares Core 1-5 Year USD Bond ETF         | Fixed Income                | LONG       |
| 9  | SPIB   | 0.0243 | State Street SPDR Portfolio Intermediat... | Investment Grade Corpora... | LONG       |
| 10 | IGIB   | 0.0243 | iShares 5-10 Year Investment Grade Corp... | Investment Grade Corpora... | LONG       |
| 14 | BSV    | 0.0242 | Vanguard Short-Term Bond ETF               | Fixed Income                | LONG       |
| 15 | VCIT   | 0.0242 | Vanguard Intermediate-Term Corporate Bo... | Investment Grade Corpora... | LONG       |
| 35 | MBB    | 0.0242 | iShares MBS ETF                            | Mortgage-Backed Fixed In... | LONG       |
| 36 | VMBS   | 0.0242 | Vanguard Mortgage-Backed Securities ETF    | Mortgage-Backed Fixed In... | LONG       |
| 5  | FBND   | 0.0241 | Fidelity Total Bond ETF                    | Active Fixed Income         | FLAT       |
| 37 | SPMB   | 0.0241 | State Street SPDR Portfolio Mortgage Ba... | Mortgage-Backed Fixed In... | LONG       |
| 24 | IGSB   | 0.0240 | iShares 1-5 Year Investment Grade Corpo... | Investment Grade Corpora... | LONG       |
| 7  | AGGY   | 0.0239 | WisdomTree Yield Enhanced U.S. Aggregat... | Aggregate Fixed Income      | LONG       |
| 26 | VCSH   | 0.0239 | Vanguard Short-Term Corporate Bond ETF     | Investment Grade Corpora... | LONG       |
| 4  | GTO    | 0.0238 | Invesco Total Return Bond ETF              | Active Fixed Income         | LONG       |
| 3  | BOND   | 0.0237 | PIMCO Active Bond Exchange-Traded Fund ... | Active Fixed Income         | LONG       |
| 25 | SLQD   | 0.0234 | iShares 0-5 Year Investment Grade Corpo... | Investment Grade Corpora... | FLAT       |
| 31 | GOVT   | 0.0233 | iShares U.S. Treasury Bond ETF             | Fixed Income                | FLAT       |
| 34 | GNMA   | 0.0233 | iShares GNMA Bond ETF                      | Mortgage-Backed Fixed In... | LONG       |
| 38 | SHY    | 0.0228 | iShares 1-3 Year Treasury Bond ETF         | Fixed Income                | LONG       |
| 28 | AGZ    | 0.0227 | iShares Agency Bond ETF                    | Agency Fixed Income         | LONG       |
| 23 | SPSB   | 0.0226 | State Street SPDR Portfolio Short Term ... | Fixed Income                | FLAT       |
| 18 | TOTL   | 0.0225 | State Street DoubleLine Total Return Ta... | Active Fixed Income         | FLAT       |
| 41 | VGSH   | 0.0225 | Vanguard Short-Term Treasury ETF           | Cash / Ultra-Short Fixed... | LONG       |
| 40 | SCHO   | 0.0221 | Schwab Short-Term U.S. Treasury ETF        | Cash / Ultra-Short Fixed... | LONG       |
| 39 | SPTS   | 0.0220 | State Street SPDR Portfolio Short Term ... | Cash / Ultra-Short Fixed... | LONG       |
| 22 | BSCQ   | 0.0214 | Invesco BulletShares 2026 Corporate Bon... | Target Maturity Bond        | FLAT       |
| 42 | BNDX   | 0.0210 | Vanguard Total International Bond ETF      | International Fixed Inco... | LONG       |
| 8  | IBDR   | 0.0208 | iShares iBonds Dec 2026 Term Corporate ... | Target Maturity Bond        | FLAT       |
| 43 | IAGG   | 0.0207 | iShares International Aggregate Bond Fu... | International Fixed Inco... | LONG       |
| 27 | CMBS   | 0.0186 | iShares CMBS Bond ETF                      | Mortgage-Backed Fixed In... | LONG       |
| 2  | LMBS   | 0.0181 | First Trust Low Duration Opportunities ... | Mortgage-Backed Fixed In... | FLAT       |
| 21 | LDUR   | 0.0171 | PIMCO Enhanced Low Duration Active Exch... | Fixed Income                | FLAT       |

**Cluster 7 – Emerging Markets Equity**

**Members:** 25   **Var explained:** 0.874   **Signal:** LONG since 2026-04-10 (36 days, 13.1%)

**Signal logic:**

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell\_cross: 2026-03-05
- Current 252d return: +49.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)

**Category breakdown**

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| Emerging Markets Equity | 19        | 0.7725     |
| Asia Equity             | 2         | 0.0813     |
| Asia ex-Japan Equity    | 1         | 0.0418     |
| Taiwan Equity           | 1         | 0.0358     |
| Global Dividend Equity  | 1         | 0.0346     |
| South Korea Equity      | 1         | 0.0340     |

**Replication basket** – IEMG (cor=0.99)   basket lead at lag 1: -0.025

| ticker | rep_wt | name                                   | category                 | etf_signal |
|--------|--------|----------------------------------------|--------------------------|------------|
| IEMG   | 0.928  | iShares Core MSCI Emerging Markets ETF | Emerging Markets Equi... | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 21 | IEMG   | 0.0425 | iShares Core MSCI Emerging Markets ETF     | Emerging Markets Equity | LONG       |
| 2  | EEM    | 0.0424 | iShares MSCI Emerging Index Fund           | Emerging Markets Equity | LONG       |
| 16 | GEM    | 0.0423 | Goldman Sachs ActiveBeta Emerging Marke... | Emerging Markets Equity | LONG       |
| 23 | SPEM   | 0.0423 | State Street SPDR Portfolio Emerging Ma... | Emerging Markets Equity | LONG       |
| 3  | VWO    | 0.0422 | Vanguard FTSE Emerging Markets ETF         | Emerging Markets Equity | FLAT       |
| 20 | ESGE   | 0.0422 | iShares ESG Aware MSCI EM ETF              | Emerging Markets Equity | LONG       |
| 22 | SCHE   | 0.0421 | Schwab Emerging Markets Equity ETF         | Emerging Markets Equity | FLAT       |
| 18 | XSOE   | 0.0420 | WisdomTree Emerging Markets Ex-State Ow... | Emerging Markets Equity | LONG       |
| 5  | AAXJ   | 0.0418 | iShares MSCI All Country Asia ex Japan ... | Asia ex-Japan Equity    | LONG       |
| 15 | EMGF   | 0.0418 | iShares Emerging Markets Equity Factor ... | Emerging Markets Equity | LONG       |
| 11 | EEMA   | 0.0413 | iShares MSCI Emerging Markets Asia ETF     | Emerging Markets Equity | LONG       |
| 14 | GMF    | 0.0411 | State Street SPDR S&P Emerging Asia Pac... | Asia Equity             | FLAT       |
| 12 | PXH    | 0.0410 | Invesco RAFI Emerging Markets ETF          | Emerging Markets Equity | FLAT       |
| 17 | FNDE   | 0.0410 | Schwab Fundamental Emerging Markets Equ... | Emerging Markets Equity | FLAT       |
| 19 | EEMV   | 0.0404 | iShares MSCI Emerging Markets Min Vol F... | Emerging Markets Equity | LONG       |
| 8  | DGS    | 0.0402 | WisdomTree Emerging Market SmallCap Fund   | Emerging Markets Equity | FLAT       |
| 10 | AIA    | 0.0401 | iShares Asia 50 ETF                        | Asia Equity             | LONG       |
| 7  | DEM    | 0.0397 | WisdomTree Emerging Markets High Divide... | Emerging Markets Equity | LONG       |
| 6  | EWX    | 0.0391 | State Street SPDR S&P Emerging Markets ... | Emerging Markets Equity | FLAT       |
| 4  | FEM    | 0.0386 | First Trust Emerging Markets AlphaDEX F... | Emerging Markets Equity | FLAT       |
| 13 | DVYE   | 0.0368 | iShares Emerging Markets Dividend Index... | Emerging Markets Equity | FLAT       |
| 9  | EWI    | 0.0358 | iShares MSCI Taiwan ETF                    | Taiwan Equity           | LONG       |
| 24 | SDIV   | 0.0346 | Global X SuperDividend ETF                 | Global Dividend Equity  | FLAT       |
| 25 | FEMS   | 0.0346 | First Trust Emerging Markets Small Cap ... | Emerging Markets Equity | LONG       |
| 1  | EWY    | 0.0340 | iShares MSCI South Korea ETF               | South Korea Equity      | LONG       |

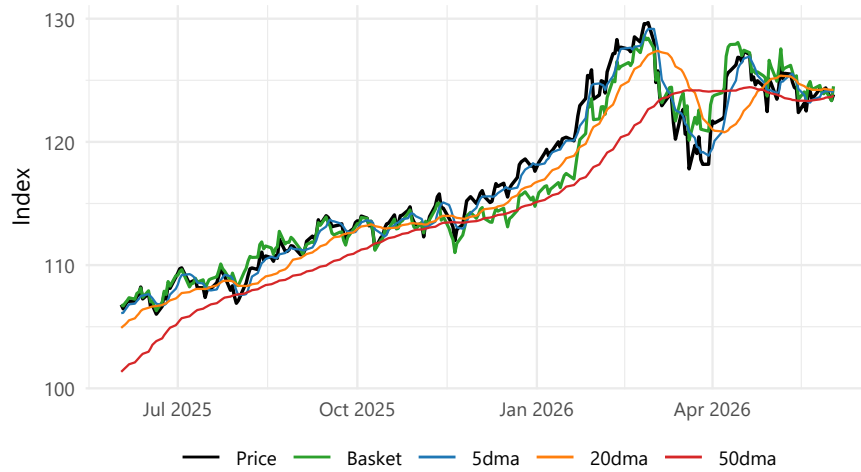
## Cluster 8 – Asia-Pacific Equity

**Members:** 23   **Var explained:** 0.685   **Signal:** LONG since 2026-05-22 (6 days, 0.3%)

### Signal logic:

- Latest buy\_cross: 2026-05-22 – filter **PASS** (252d at date = +17.2%)

- Latest sell\_cross: 2026-05-18
- Current 252d return: +17.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-22 > last sell at 2026-05-18)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| Asia-Pacific Equity           | 2         | 0.0979     |
| International Dividend Equity | 2         | 0.0960     |
| International Equity          | 2         | 0.0959     |
| Global Equity                 | 2         | 0.0921     |
| Real Estate                   | 2         | 0.0917     |
| Emerging Markets Fixed Income | 2         | 0.0837     |
| Currency                      | 2         | 0.0814     |
| Australia Equity              | 1         | 0.0486     |
| Canada Equity                 | 1         | 0.0462     |
| Switzerland Equity            | 1         | 0.0431     |
| South Africa Equity           | 1         | 0.0410     |
| Latin America Equity          | 1         | 0.0395     |
| New Zealand Equity            | 1         | 0.0394     |
| Mexico Equity                 | 1         | 0.0373     |
| Brazil Equity                 | 1         | 0.0334     |
| Chile Equity                  | 1         | 0.0331     |

**Replication basket** – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.002

| ticker | rep_wt | name                                     | category            | etf_signal |
|--------|--------|------------------------------------------|---------------------|------------|
| EWZ    | 0.094  | iShares MSCI Brazil ETF                  | Brazil Equity       | FLAT       |
| IGF    | 0.289  | iShares Global Infrastructure ETF        | Global Equity       | FLAT       |
| BBAX   | 0.453  | JPMorgan BetaBuilders Developed Asia ... | Asia-Pacific Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

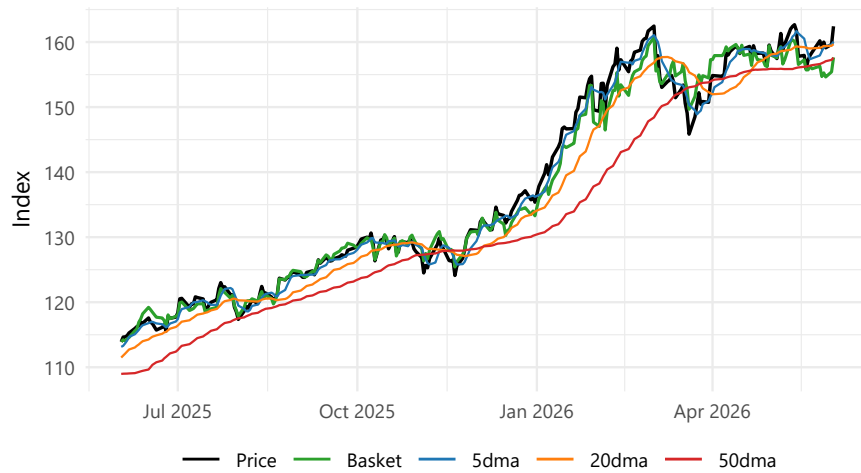
|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 7  | EPP    | 0.0489 | iShares MSCI Pacific Ex-Japan Index Fund   | Asia-Pacific Equity         | FLAT       |
| 11 | IDLV   | 0.0489 | Invesco S&P International Developed Low... | International Equity        | FLAT       |
| 23 | BBAX   | 0.0489 | JPMorgan BetaBuilders Developed Asia Pa... | Asia-Pacific Equity         | FLAT       |
| 4  | EWA    | 0.0486 | iShares MSCI Australia Index Fund          | Australia Equity            | LONG       |
| 10 | DWX    | 0.0483 | State Street SPDR S&P International Div... | International Dividend E... | LONG       |
| 3  | PID    | 0.0478 | Invesco International Dividend Achiever... | International Dividend E... | LONG       |
| 12 | EFAV   | 0.0470 | iShares MSCI EAFE Min Vol Factor ETF       | International Equity        | FLAT       |
| 8  | NFRA   | 0.0467 | FlexShares STOXX Global Broad Infrastru... | Global Equity               | LONG       |
| 13 | VNQI   | 0.0464 | Vanguard Global ex-U.S. Real Estate ETF    | Real Estate                 | FLAT       |
| 6  | EWC    | 0.0462 | iShares MSCI Canada Index Fund             | Canada Equity               | LONG       |
| 9  | IGF    | 0.0454 | iShares Global Infrastructure ETF          | Global Equity               | FLAT       |
| 2  | RWX    | 0.0453 | State Street SPDR Dow Jones Internation... | Real Estate                 | FLAT       |
| 5  | EWL    | 0.0431 | iShares MSCI Switzerland ETF               | Switzerland Equity          | LONG       |
| 19 | FXA    | 0.0431 | Invesco CurrencyShares Australian Dolla... | Currency                    | FLAT       |
| 14 | EBND   | 0.0419 | State Street SPDR Bloomberg Emerging Ma... | Emerging Markets Fixed I... | LONG       |
| 15 | EMLC   | 0.0417 | VanEck J. P. Morgan EM Local Currency B... | Emerging Markets Fixed I... | LONG       |
| 21 | EZA    | 0.0410 | iShares MSCI South Africa Index Fund       | South Africa Equity         | FLAT       |
| 18 | ILF    | 0.0395 | iShares Latin America 40 ETF               | Latin America Equity        | FLAT       |
| 20 | ENZL   | 0.0394 | iShares MSCI New Zealand ETF               | New Zealand Equity          | LONG       |
| 22 | FXC    | 0.0382 | Invesco CurrencyShares Canadian Dollar ... | Currency                    | FLAT       |
| 16 | EWX    | 0.0373 | iShares MSCI Mexico ETF                    | Mexico Equity               | LONG       |
| 1  | EWZ    | 0.0334 | iShares MSCI Brazil ETF                    | Brazil Equity               | FLAT       |
| 17 | ECH    | 0.0331 | iShares MSCI Chile ETF                     | Chile Equity                | LONG       |

## Cluster 9 – Materials

**Members:** 21 **Var explained:** 0.705 **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-04-07 – filter **PASS** (252d at date = +44.3%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +44.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-07)



### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Materials                   | 13        | 0.6532     |
| MLP / Energy Infrastructure | 4         | 0.1703     |
| Energy                      | 1         | 0.0501     |
| Consumer Staples            | 1         | 0.0493     |
| Norway Equity               | 1         | 0.0439     |
| Industrial Metals           | 1         | 0.0333     |

**Replication basket** – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.042

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| AMLP   | 0.114  | Alerian MLP ETF                          | MLP / Energy Infrastr... | FLAT       |
| URA    | 0.081  | Global X Uranium ETF                     | Materials                | FLAT       |
| GUNR   | 0.773  | FlexShares Global Upstream Natural Re... | Materials                | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 15 | GNR    | 0.0549 | State Street SPDR S&P Global Natural Re... | Materials                   | FLAT       |
| 14 | GUNR   | 0.0542 | FlexShares Global Upstream Natural Reso... | Materials                   | FLAT       |
| 10 | MXI    | 0.0528 | iShares Global Materials ETF               | Materials                   | FLAT       |
| 11 | IYM    | 0.0528 | iShares U.S. Basic Materials ETF           | Materials                   | LONG       |
| 21 | NANR   | 0.0523 | State Street SPDR S&P North American Na... | Materials                   | FLAT       |
| 13 | VAW    | 0.0513 | Vanguard Materials ETF                     | Materials                   | LONG       |
| 16 | PICK   | 0.0508 | iShares MSCI Global Select Metals & Min... | Materials                   | FLAT       |
| 12 | RSPM   | 0.0506 | Invesco S&P 500 Equal Weight Materials ... | Materials                   | LONG       |
| 1  | XLB    | 0.0504 | State Street Materials Select Sector SP... | Materials                   | LONG       |
| 20 | IGE    | 0.0501 | iShares North American Natural Resource... | Energy                      | FLAT       |
| 4  | XME    | 0.0498 | State Street SPDR S&P Metals & Mining E... | Materials                   | LONG       |
| 17 | SLX    | 0.0496 | VanEck Steel ETF                           | Materials                   | LONG       |
| 6  | MOO    | 0.0493 | VanEck Agribusiness ETF                    | Consumer Staples            | FLAT       |
| 3  | COPX   | 0.0472 | Global X Copper Miners ETF                 | Materials                   | LONG       |
| 5  | EMLP   | 0.0441 | First Trust North American Energy Infra... | MLP / Energy Infrastruct... | FLAT       |
| 9  | MLPX   | 0.0439 | Global X MLP & Energy Infrastructure ETF   | MLP / Energy Infrastruct... | FLAT       |
| 18 | NORW   | 0.0439 | Global X MSCI Norway ETF                   | Norway Equity               | FLAT       |
| 7  | AMLP   | 0.0417 | Alerian MLP ETF                            | MLP / Energy Infrastruct... | FLAT       |
| 8  | MLPA   | 0.0406 | Global X MLP ETF                           | MLP / Energy Infrastruct... | FLAT       |
| 2  | URA    | 0.0365 | Global X Uranium ETF                       | Materials                   | FLAT       |
| 19 | DBB    | 0.0333 | Invesco DB Base Metals Fund                | Industrial Metals           | LONG       |

## Cluster 10 – High Yield Bond

**Members:** 21   **Var explained:** 0.766   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell\_cross: 2026-05-18
- Current 252d return: +8.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-13)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| High Yield Bond               | 10        | 0.5154     |
| Preferred Stock               | 7         | 0.3062     |
| Emerging Markets Fixed Income | 4         | 0.1783     |

**Replication basket** – HYG (cor=0.95) basket lead at lag 1: 0.049

| ticker | rep_wt | name                                      | category        | etf_signal |
|--------|--------|-------------------------------------------|-----------------|------------|
| HYG    | 0.916  | iShares iBoxx \$ High Yield Corporate ... | High Yield Bond | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                        | category        | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------|------------|
| 13 | JNK    | 0.0526 | State Street SPDR Bloomberg High Yield ...  | High Yield Bond | FLAT       |
| 1  | HYG    | 0.0525 | iShares iBoxx \$ High Yield Corporate Bo... | High Yield Bond | FLAT       |
| 19 | HYLB   | 0.0524 | Xtrackers USD High Yield Corporate Bond...  | High Yield Bond | FLAT       |
| 18 | BBHY   | 0.0523 | JPMorgan BetaBuilders USD High Yield Co...  | High Yield Bond | FLAT       |
| 16 | SHYG   | 0.0521 | iShares 0-5 Year High Yield Corporate B...  | High Yield Bond | FLAT       |



|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 14 | SJNK | 0.0520 | State Street SPDR Bloomberg Short Term ... | High Yield Bond             | FLAT |
| 15 | ANGL | 0.0516 | VanEck Fallen Angel High Yield Bond ETF    | High Yield Bond             | FLAT |
| 17 | FALN | 0.0516 | iShares Fallen Angels USD Bond ETF         | High Yield Bond             | FLAT |
| 6  | HYS  | 0.0511 | PIMCO 0-5 Year High Yield Corporat Bond... | High Yield Bond             | FLAT |
| 9  | PFF  | 0.0480 | iShares Preferred and Income Securities... | Preferred Stock             | FLAT |
| 3  | HYLS | 0.0473 | First Trust Tactical High Yield ETF        | High Yield Bond             | FLAT |
| 2  | EMB  | 0.0467 | iShares J.P. Morgan USD Emerging Market... | Emerging Markets Fixed I... | FLAT |
| 21 | VWOB | 0.0466 | Vanguard Emerging Markets Government Bo... | Emerging Markets Fixed I... | LONG |
| 11 | PFXF | 0.0463 | VanEck Preferred Securities ex Financia... | Preferred Stock             | FLAT |
| 20 | PCY  | 0.0457 | Invesco Emerging Markets Sovereign Debt... | Emerging Markets Fixed I... | LONG |
| 7  | PGX  | 0.0446 | Invesco Preferred ETF                      | Preferred Stock             | FLAT |
| 10 | PSK  | 0.0441 | State Street SPDR ICE Preferred Securit... | Preferred Stock             | FLAT |
| 4  | FPE  | 0.0433 | First Trust Preferred Securities and In... | Preferred Stock             | LONG |
| 5  | PGF  | 0.0431 | Invesco Financial Preferred ETF            | Preferred Stock             | FLAT |
| 12 | HYEM | 0.0393 | VanEck Emerging Markets High Yield Bond... | Emerging Markets Fixed I... | LONG |
| 8  | VRP  | 0.0368 | Invesco Variable Rate Preferred ETF        | Preferred Stock             | FLAT |

## Cluster 11 – Healthcare

**Members:** 17   **Var explained:** 0.762   **Signal:** LONG since 2026-05-06 (18 days, -1.3%)

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell\_cross: 2026-04-29
- Current 252d return: +15.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Healthcare                | 16        | 0.9419     |
| U.S. Small Cap Healthcare | 1         | 0.0581     |

**Replication basket** – VHT (cor=0.93) basket lead at lag 1: -0.002

| ticker | rep_wt | name                     | category   | etf_signal |
|--------|--------|--------------------------|------------|------------|
| VHT    | 1.101  | Vanguard Health Care ETF | Healthcare | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 15 | VHT    | 0.0653 | Vanguard Health Care ETF                   | Healthcare | LONG       |
| 16 | FHLC   | 0.0653 | Fidelity MSCI Health Care Index ETF        | Healthcare | LONG       |
| 9  | FXH    | 0.0646 | First Trust Health Care AlphaDEX           | Healthcare | LONG       |
| 14 | RSPH   | 0.0640 | Invesco S&P 500 Equal Weight Health Car... | Healthcare | LONG       |
| 13 | IYH    | 0.0637 | iShares U.S. Healthcare ETF                | Healthcare | LONG       |
| 10 | IXJ    | 0.0623 | iShares Global Healthcare ETF              | Healthcare | LONG       |

|    |      |        |                                            |                           |      |
|----|------|--------|--------------------------------------------|---------------------------|------|
| 1  | XLV  | 0.0621 | State Street Health Care Select Sector ... | Healthcare                | LONG |
| 3  | IBB  | 0.0605 | iShares Biotechnology ETF                  | Healthcare                | LONG |
| 6  | FBT  | 0.0596 | First Trust Amex Biotech Index Fund        | Healthcare                | FLAT |
| 8  | PSCH | 0.0581 | Invesco S&P SmallCap Health Care ETF       | U.S. Small Cap Healthcare | LONG |
| 7  | XHE  | 0.0574 | State Street SPDR S&P Health Care Equip... | Healthcare                | LONG |
| 12 | IHI  | 0.0571 | iShares U.S. Medical Devices ETF           | Healthcare                | FLAT |
| 5  | PTH  | 0.0544 | Invesco Dorsey Wright Healthcare Moment... | Healthcare                | FLAT |
| 17 | PPH  | 0.0543 | VanEck Pharmaceutical ETF                  | Healthcare                | LONG |
| 4  | XBI  | 0.0535 | State Street SPDR S&P Biotech ETF          | Healthcare                | FLAT |
| 2  | ARKG | 0.0510 | ARK Genomic Revolution ETF                 | Healthcare                | FLAT |
| 11 | IHF  | 0.0470 | iShares U.S. Health Care Providers ETF     | Healthcare                | LONG |

## Cluster 12 – China Equity

**Members:** 15   **Var explained:** 0.803   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +17.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



**Category breakdown**

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| China Equity            | 13        | 0.8766     |
| Emerging Markets Equity | 1         | 0.0642     |
| Hong Kong Equity        | 1         | 0.0591     |

**Replication basket** – KWEB (cor=0.92) basket lead at lag 1: 0.011

| ticker | rep_wt | name                               | category     | etf_signal |
|--------|--------|------------------------------------|--------------|------------|
| KWEB   | 0.563  | KraneShares CSI China Internet ETF | China Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

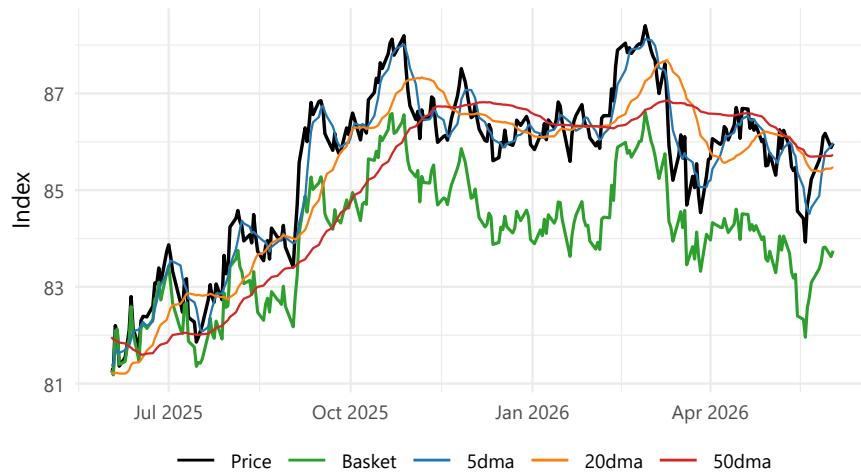
|    | ticker | PC_wt  | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 11 | CXSE   | 0.0737 | WisdomTree China ex-State-Owned Enterpr... | China Equity            | FLAT       |
| 9  | GXC    | 0.0735 | State Street SPDR S&P China ETF            | China Equity            | FLAT       |
| 1  | MCHI   | 0.0729 | iShares MSCI China ETF                     | China Equity            | FLAT       |
| 2  | FXI    | 0.0712 | iShares China Large-Cap ETF                | China Equity            | FLAT       |
| 5  | CQQQ   | 0.0712 | Invesco China Technology ETF               | China Equity            | LONG       |
| 7  | CHIQ   | 0.0706 | Global X MSCI China Consumer Discretion... | China Equity            | FLAT       |
| 4  | KWEB   | 0.0679 | KraneShares CSI China Internet ETF         | China Equity            | FLAT       |
| 6  | ASHR   | 0.0669 | Xtrackers Harvest CSI 300 China A-Share... | China Equity            | FLAT       |
| 8  | CNYA   | 0.0661 | iShares MSCI China A ETF                   | China Equity            | FLAT       |
| 13 | KGRN   | 0.0651 | KraneShares MSCI China Clean Technology... | China Equity            | FLAT       |
| 3  | EMQQ   | 0.0642 | EMQQ The Emerging Markets Internet ETF     | Emerging Markets Equity | FLAT       |
| 10 | KBA    | 0.0595 | KraneShares Boseri MSCI China A 50 Conn... | China Equity            | FLAT       |
| 14 | CNXT   | 0.0595 | VanEck ChiNext Innovators ETF              | China Equity            | LONG       |
| 12 | EWK    | 0.0591 | iShares MSCI Hong Kong Index Fund          | Hong Kong Equity        | FLAT       |
| 15 | KURE   | 0.0586 | KraneShares MSCI All China Health Care ... | China Equity            | FLAT       |

**Cluster 13 – Fixed Income**

**Members:** 15 **Var explained:** 0.928 **Signal:** LONG since 2026-05-29 (2 days, -0.2%)

**Signal logic:**

- Latest buy\_cross: 2026-05-29 – filter **PASS** (252d at date = +6.4%)
- Latest sell\_cross: 2026-05-13
- Current 252d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-13)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Investment Grade Corporate Bond | 7         | 0.4706     |
| Fixed Income                    | 6         | 0.4009     |
| Active Fixed Income             | 1         | 0.0665     |
| Taxable Municipal Bond          | 1         | 0.0619     |

**Replication basket** – TLT (cor=0.97) basket lead at lag 1: -0.041

| ticker | rep_wt | name                               | category     | etf_signal |
|--------|--------|------------------------------------|--------------|------------|
| TLT    | 0.694  | iShares 20+ Year Treasury Bond ETF | Fixed Income | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

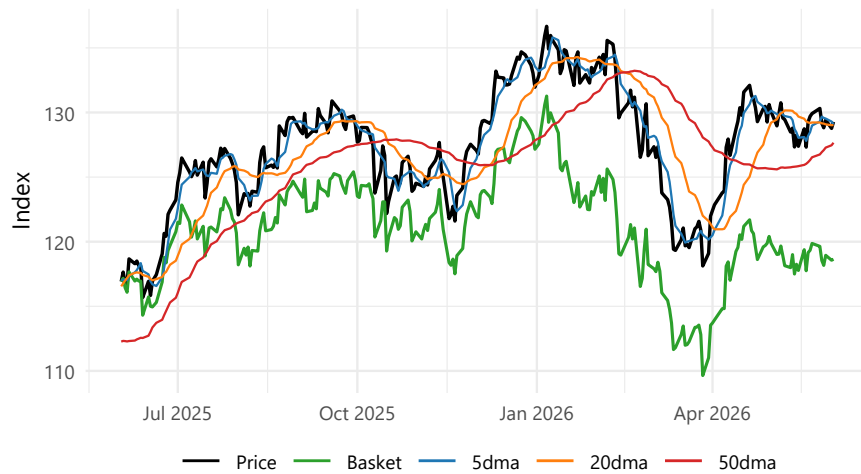
|    | ticker | PC_wt  | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 9  | BLV    | 0.0687 | Vanguard Long-Term Bond ETF                 | Fixed Income                | LONG       |
| 7  | SPLB   | 0.0678 | State Street SPDR Portfolio Long Term C...  | Investment Grade Corpora... | LONG       |
| 8  | IGLB   | 0.0678 | iShares 10 Year Investment Grade Corpor...  | Investment Grade Corpora... | LONG       |
| 10 | VCLT   | 0.0676 | Vanguard Long-Term Corporate Bond ETF       | Investment Grade Corpora... | LONG       |
| 2  | LQD    | 0.0675 | iShares iBoxx \$ Investment Grade Corpor... | Investment Grade Corpora... | LONG       |
| 11 | USIG   | 0.0672 | iShares Broad USD Investment Grade Corp...  | Investment Grade Corpora... | LONG       |
| 6  | QLTA   | 0.0671 | iShares Aaa A Rated Corporate Bond ETF      | Investment Grade Corpora... | LONG       |
| 12 | TLH    | 0.0668 | iShares 10-20 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 14 | SPTL   | 0.0667 | State Street SPDR Portfolio Long Term T...  | Fixed Income                | FLAT       |
| 15 | VGLT   | 0.0666 | Vanguard Long-Term Treasury ETF             | Fixed Income                | FLAT       |
| 3  | FLCO   | 0.0665 | Franklin Investment Grade Corporate ETF     | Active Fixed Income         | LONG       |
| 1  | TLT    | 0.0664 | iShares 20+ Year Treasury Bond ETF          | Fixed Income                | FLAT       |
| 5  | CORP   | 0.0657 | PIMCO Investment Grade Corporate Bond I...  | Investment Grade Corpora... | LONG       |
| 13 | EDV    | 0.0656 | Vanguard Extended Duration Treasury ETF     | Fixed Income                | LONG       |
| 4  | BAB    | 0.0619 | Invesco Taxable Municipal Bond ETF          | Taxable Municipal Bond      | FLAT       |

## Cluster 14 – Financials

**Members:** 14   **Var explained:** 0.877   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell\_cross: 2026-05-11
- Current 252d return: +10.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



### Category breakdown

| category      | n_members | abs_weight |
|---------------|-----------|------------|
| Financials    | 13        | 0.938      |
| Global Equity | 1         | 0.062      |

**Replication basket** – XLF (cor=0.96) basket lead at lag 1: 0.014

| ticker | rep_wt | name                                     | category   | etf_signal |
|--------|--------|------------------------------------------|------------|------------|
| XLF    | 1.097  | State Street Financial Select Sector ... | Financials | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 10 | VFH    | 0.0754 | Vanguard Financials ETF                    | Financials | FLAT       |
| 11 | FNCL   | 0.0754 | Fidelity MSCI Financials Index ETF         | Financials | LONG       |
| 5  | FXO    | 0.0751 | First Trust Financials AlphaDEX            | Financials | FLAT       |
| 7  | IYF    | 0.0751 | iShares U.S. Financial ETF                 | Financials | FLAT       |
| 8  | IYG    | 0.0747 | iShares U.S. Financial Services ETF        | Financials | FLAT       |
| 1  | XLF    | 0.0740 | State Street Financial Select Sector SP... | Financials | LONG       |

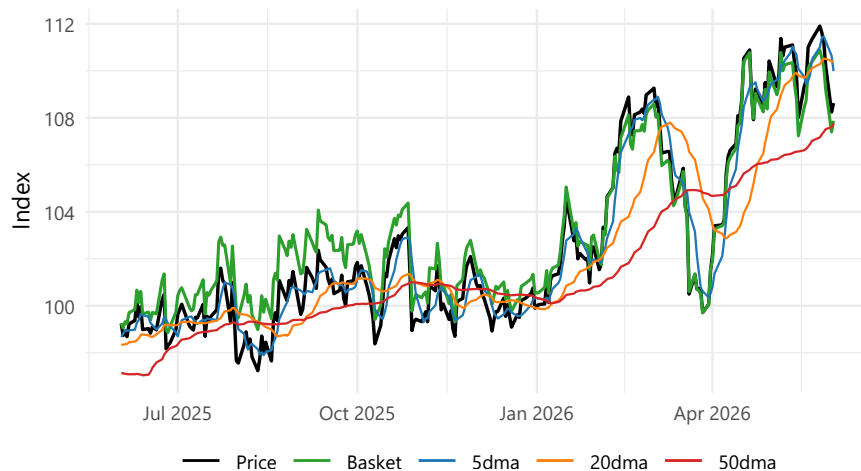
|    |      |        |                                            |               |      |
|----|------|--------|--------------------------------------------|---------------|------|
| 9  | KBWB | 0.0732 | Invesco KBW Bank ETF                       | Financials    | FLAT |
| 3  | KBE  | 0.0712 | State Street SPDR S&P Bank ETF             | Financials    | FLAT |
| 14 | KCE  | 0.0708 | State Street SPDR S&P Capital Markets E... | Financials    | LONG |
| 6  | IAT  | 0.0705 | iShares U.S. Regional Banks ETF            | Financials    | LONG |
| 13 | IAI  | 0.0699 | iShares U.S. Broker-Dealers & Securitie... | Financials    | LONG |
| 2  | KRE  | 0.0687 | State Street SPDR S&P Regional Banking ... | Financials    | FLAT |
| 4  | KIE  | 0.0640 | State Street SPDR S&P Insurance ETF        | Financials    | FLAT |
| 12 | PSP  | 0.0620 | Invesco Global Listed Private Equity ETF   | Global Equity | FLAT |

### Cluster 15 – Real Estate

**Members:** 14   **Var explained:** 0.962   **Signal:** LONG since 2026-04-10 (36 days, 1.9%)

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell\_cross: 2026-03-10
- Current 252d return: +9.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



### Category breakdown



| category    | n_members | abs_weight |
|-------------|-----------|------------|
| Real Estate | 14        | 1          |

**Replication basket** – VNQ (cor=0.99) basket lead at lag 1: 0.002

| ticker | rep_wt | name                     | category    | etf_signal |
|--------|--------|--------------------------|-------------|------------|
| VNQ    | 0.971  | Vanguard Real Estate ETF | Real Estate | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

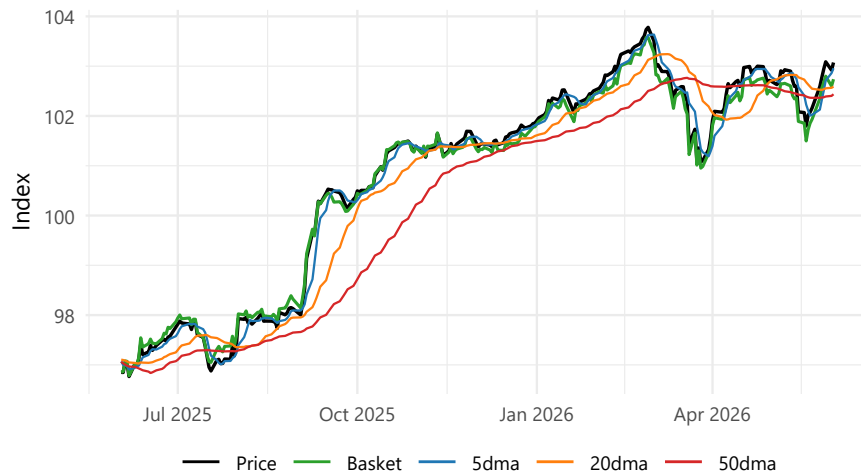
|    | ticker | PC_wt  | name                                       | category    | etf_signal |
|----|--------|--------|--------------------------------------------|-------------|------------|
| 8  | VNQ    | 0.0725 | Vanguard Real Estate ETF                   | Real Estate | FLAT       |
| 9  | FREL   | 0.0725 | Fidelity MSCI Real Estate Index ETF        | Real Estate | FLAT       |
| 11 | SCHH   | 0.0725 | Schwab U.S. REIT ETF                       | Real Estate | LONG       |
| 2  | IYR    | 0.0724 | iShares U.S. Real Estate ETF               | Real Estate | FLAT       |
| 10 | USRT   | 0.0724 | iShares Core U.S. REIT ETF                 | Real Estate | LONG       |
| 6  | RWR    | 0.0723 | State Street SPDR Dow Jones REIT ETF       | Real Estate | LONG       |
| 13 | PSR    | 0.0721 | Invesco Active U.S. Real Estate Fund       | Real Estate | LONG       |
| 1  | XLRE   | 0.0719 | State Street Real Estate Select Sector ... | Real Estate | FLAT       |
| 5  | ICF    | 0.0719 | iShares Select U.S. REIT ETF               | Real Estate | LONG       |
| 7  | REET   | 0.0715 | iShares Global REIT ETF                    | Real Estate | LONG       |
| 3  | RWO    | 0.0711 | State Street SPDR Dow Jones Global Real... | Real Estate | FLAT       |
| 4  | REZ    | 0.0698 | iShares Residential and Multisector Rea... | Real Estate | FLAT       |
| 14 | NURE   | 0.0688 | Nuveen Short-Term REIT ETF                 | Real Estate | LONG       |
| 12 | INDS   | 0.0683 | Pacer Industrial Real Estate ETF           | Real Estate | FLAT       |

## Cluster 16 – Municipal Bond

**Members:** 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (3 days, 0.1%)

### Signal logic:

- Latest buy\_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell\_cross: 2026-05-15
- Current 252d return: +6.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



### Category breakdown

| category       | n_members | abs_weight |
|----------------|-----------|------------|
| Municipal Bond | 13        | 1          |

**Replication basket** – MUB (cor=0.95) basket lead at lag 1: 0.181

| ticker | rep_wt | name                           | category       | etf_signal |
|--------|--------|--------------------------------|----------------|------------|
| MUB    | 0.905  | iShares National Muni Bond ETF | Municipal Bond | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category       | etf_signal |
|----|--------|--------|--------------------------------------------|----------------|------------|
| 13 | VTEB   | 0.0817 | Vanguard Tax-Exempt Bond ETF               | Municipal Bond | LONG       |
| 11 | MUB    | 0.0813 | iShares National Muni Bond ETF             | Municipal Bond | LONG       |
| 6  | CMF    | 0.0801 | iShares California Muni Bond ETF           | Municipal Bond | LONG       |
| 9  | TFI    | 0.0801 | State Street SPDR Nuveen ICE Municipal ... | Municipal Bond | LONG       |
| 8  | ITM    | 0.0800 | VanEck Intermediate Muni ETF               | Municipal Bond | LONG       |
| 1  | FMB    | 0.0795 | First Trust Managed Municipal ETF          | Municipal Bond | LONG       |
| 7  | NYF    | 0.0793 | iShares New York Muni Bond ETF             | Municipal Bond | LONG       |

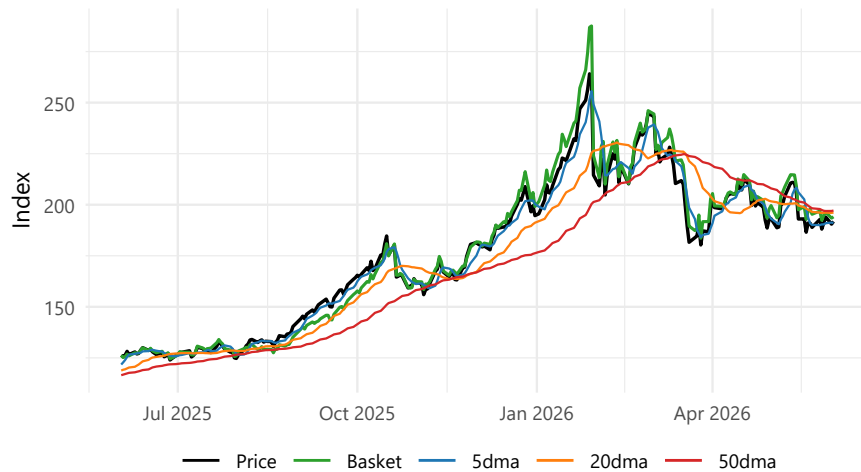
|    |      |        |                                            |                |      |
|----|------|--------|--------------------------------------------|----------------|------|
| 5  | PZA  | 0.0780 | Invesco National AMT-Free Municipal Bon... | Municipal Bond | LONG |
| 4  | MUNI | 0.0769 | PIMCO Intermediate Municipal Bond Activ... | Municipal Bond | LONG |
| 2  | HYD  | 0.0748 | VanEck High Yield Muni ETF                 | Municipal Bond | LONG |
| 3  | HYMB | 0.0720 | State Street SPDR Nuveen ICE High Yield... | Municipal Bond | LONG |
| 12 | SUB  | 0.0695 | iShares Short-Term National Muni Bond E... | Municipal Bond | LONG |
| 10 | SHM  | 0.0667 | State Street SPDR Nuveen ICE Short Term... | Municipal Bond | LONG |

## Cluster 17 – Gold

**Members:** 12   **Var explained:** 0.858   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +58.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
|----------|-----------|------------|

|                 |   |        |
|-----------------|---|--------|
| Materials       | 5 | 0.4157 |
| Gold            | 3 | 0.2502 |
| Precious Metals | 2 | 0.1725 |
| Silver          | 2 | 0.1616 |

**Replication basket** – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.014

| ticker | rep_wt | name                      | category | etf_signal |
|--------|--------|---------------------------|----------|------------|
| GLD    | 0.042  | SPDR Gold Shares          | Gold     | FLAT       |
| IAU    | 0.729  | iShares Gold Trust Shares | Gold     | FLAT       |
| SLV    | 0.410  | iShares Silver Trust      | Silver   | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category        | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------|------------|
| 11 | DBP    | 0.0866 | Invesco DB Precious Metals Fund            | Precious Metals | FLAT       |
| 7  | GLTR   | 0.0859 | abrdn Physical Precious Metals Basket S... | Precious Metals | FLAT       |
| 1  | GDX    | 0.0840 | VanEck Gold Miners ETF                     | Materials       | FLAT       |
| 2  | GDXJ   | 0.0840 | VanEck Junior Gold Miners ETF              | Materials       | FLAT       |
| 3  | GLD    | 0.0834 | SPDR Gold Shares                           | Gold            | FLAT       |
| 9  | IAU    | 0.0834 | iShares Gold Trust Shares                  | Gold            | FLAT       |
| 10 | SGOL   | 0.0833 | abrdn Physical Gold Shares ETF             | Gold            | FLAT       |
| 12 | GOAU   | 0.0832 | US Global GO Gold and Precious Metal Mi... | Materials       | FLAT       |
| 6  | SIL    | 0.0828 | Global X Silver Miners ETF                 | Materials       | FLAT       |
| 5  | SILJ   | 0.0817 | Amplify Junior Silver Miners ETF           | Materials       | LONG       |
| 4  | SLV    | 0.0809 | iShares Silver Trust                       | Silver          | LONG       |
| 8  | SIVR   | 0.0807 | abrdn Physical Silver Shares ETF           | Silver          | LONG       |

## Cluster 18 – Consumer Staples

**Members:** 11 **Var explained:** 0.745 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-22 – filter **PASS** (252d at date = +7.3%)
- Latest sell\_cross: 2026-05-29
- Current 252d return: +2.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-29 more recent than last effective buy 2026-05-22)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Consumer Staples                | 6         | 0.5732     |
| Utilities                       | 4         | 0.3522     |
| U.S. Small Cap Consumer Staples | 1         | 0.0746     |

**Replication basket** – XLU+XLP (cor=0.99) basket lead at lag 1: -0.008

| ticker | rep_wt | name                                     | category         | etf_signal |
|--------|--------|------------------------------------------|------------------|------------|
| XLU    | 0.364  | State Street Utilities Select Sector ... | Utilities        | FLAT       |
| XLP    | 0.587  | State Street Consumer Staples Select ... | Consumer Staples | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category         | etf_signal |
|---|--------|--------|--------------------------------------------|------------------|------------|
| 5 | VDC    | 0.0983 | Vanguard Consumer Staples ETF              | Consumer Staples | FLAT       |
| 6 | FSTA   | 0.0983 | Fidelity MSCI Consumer Staples Index ETF   | Consumer Staples | FLAT       |
| 1 | XLP    | 0.0974 | State Street Consumer Staples Select Se... | Consumer Staples | FLAT       |
| 4 | IYK    | 0.0949 | iShares U.S. Consumer Staples ETF          | Consumer Staples | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 3  | KXI  | 0.0946 | iShares Global Consumer Staples ETF        | Consumer Staples            | FLAT |
| 11 | PBJ  | 0.0896 | Invesco Food & Beverage ETF                | Consumer Staples            | FLAT |
| 7  | IDU  | 0.0891 | iShares U.S. Utilities ETF                 | Utilities                   | FLAT |
| 8  | VPU  | 0.0879 | Vanguard Utilities ETF                     | Utilities                   | FLAT |
| 9  | FUTY | 0.0879 | Fidelity MSCI Utilities Index ETF          | Utilities                   | FLAT |
| 2  | XLU  | 0.0874 | State Street Utilities Select Sector SP... | Utilities                   | FLAT |
| 10 | PSCC | 0.0746 | Invesco S&P SmallCap Consumer Staples E... | U.S. Small Cap Consumer ... | FLAT |

## Cluster 19 – India Equity

Members: 11 Var explained: 0.370 Signal: FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (252d at date = +4.3%)
- Latest sell\_cross: 2026-05-13
- Current 252d return: -11.3% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



### Category breakdown

| category             | n_members | abs_weight |
|----------------------|-----------|------------|
| India Equity         | 3         | 0.4645     |
| Malaysia Equity      | 1         | 0.1127     |
| Indonesia Equity     | 1         | 0.1060     |
| Philippines Equity   | 1         | 0.1011     |
| Vietnam Equity       | 1         | 0.0743     |
| Turkey Equity        | 1         | 0.0598     |
| U.S. Dividend Equity | 1         | 0.0547     |
| Platinum             | 1         | 0.0226     |
| Natural Gas          | 1         | 0.0044     |

**Replication basket** – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.004

| ticker | rep_wt | name                                     | category             | etf_signal |
|--------|--------|------------------------------------------|----------------------|------------|
| NOBL   | 0.080  | ProShares S&P 500 Dividend Aristocrat... | U.S. Dividend Equity | FLAT       |
| INDA   | 0.656  | iShares MSCI India ETF                   | India Equity         | FLAT       |
| PPLT   | 0.028  | abrdn Physical Platinum Shares ETF       | Platinum             | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

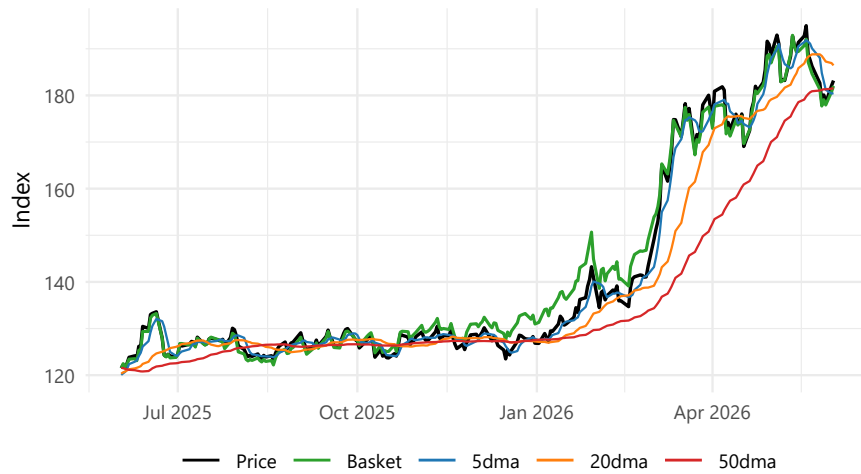
|    | ticker | PC_wt  | name                                       | category             | etf_signal |
|----|--------|--------|--------------------------------------------|----------------------|------------|
| 1  | INDA   | 0.1561 | iShares MSCI India ETF                     | India Equity         | FLAT       |
| 2  | INDY   | 0.1545 | iShares India 50 ETF                       | India Equity         | FLAT       |
| 3  | EPI    | 0.1538 | WisdomTree India Earnings Fund             | India Equity         | FLAT       |
| 9  | EWM    | 0.1127 | iShares MSCI Malaysia Index Fund           | Malaysia Equity      | FLAT       |
| 7  | EIDO   | 0.1060 | iShares MSCI Indonesia ETF                 | Indonesia Equity     | FLAT       |
| 8  | EPHE   | 0.1011 | iShares MSCI Philippines ETF               | Philippines Equity   | FLAT       |
| 11 | VNM    | 0.0743 | VanEck Vietnam ETF                         | Vietnam Equity       | FLAT       |
| 10 | TUR    | 0.0598 | iShares MSCI Turkey ETF                    | Turkey Equity        | FLAT       |
| 4  | NOBL   | 0.0547 | ProShares S&P 500 Dividend Aristocrats ... | U.S. Dividend Equity | FLAT       |
| 5  | PPLT   | 0.0226 | abrdn Physical Platinum Shares ETF         | Platinum             | FLAT       |
| 6  | UNG    | 0.0044 | United States Natural Gas Fund LP          | Natural Gas          | FLAT       |

## Cluster 20 – Broad Commodities

**Members:** 10 **Var explained:** 0.841 **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-01-08 – filter **PASS** (252d at date = +2.8%)
- Latest sell\_cross: 2026-05-27
- Current 252d return: +54.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-27 more recent than last effective buy 2026-01-08)



### Category breakdown

| category                 | n_members | abs_weight |
|--------------------------|-----------|------------|
| Broad Commodities        | 6         | 0.6371     |
| Oil                      | 2         | 0.2056     |
| Energy Commodities       | 1         | 0.1053     |
| Agricultural Commodities | 1         | 0.0519     |

**Replication basket** – PDBC (cor=0.98) basket lead at lag 1: 0.074

| ticker | rep_wt | name                                     | category          | etf_signal |
|--------|--------|------------------------------------------|-------------------|------------|
| PDBC   | 1.129  | Invesco Optimum Yield Diversified Com... | Broad Commodities | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)



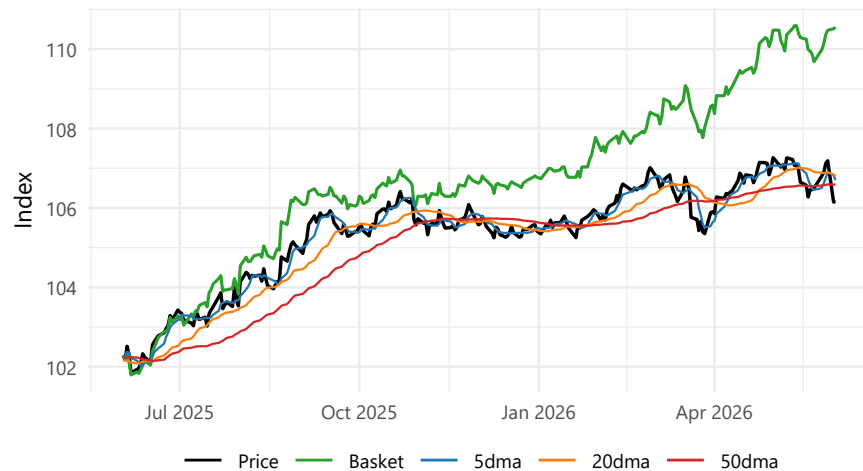
|    | ticker | PC_wt  | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 2  | DBC    | 0.1091 | Invesco DB Commodity Index Tracking Fund   | Broad Commodities        | FLAT       |
| 6  | GSG    | 0.1091 | iShares GSCI Commodity-Indexed Trust Fu... | Broad Commodities        | FLAT       |
| 8  | PDBC   | 0.1090 | Invesco Optimum Yield Diversified Commo... | Broad Commodities        | FLAT       |
| 9  | COMT   | 0.1089 | iShares GSCI Commodity Dynamic Roll Str... | Broad Commodities        | FLAT       |
| 10 | DBE    | 0.1053 | Invesco DB Energy Fund                     | Energy Commodities       | FLAT       |
| 1  | USO    | 0.1033 | United States Oil Fund                     | Oil                      | FLAT       |
| 5  | DBO    | 0.1022 | Invesco DB Oil Fund                        | Oil                      | FLAT       |
| 7  | DJP    | 0.1006 | iPath Bloomberg Commodity Index Total R... | Broad Commodities        | FLAT       |
| 4  | FTGC   | 0.1004 | First Trust Global Tactical Commodity S... | Broad Commodities        | FLAT       |
| 3  | DBA    | 0.0519 | Invesco DB Agriculture Fund                | Agricultural Commodities | FLAT       |

### Cluster 21 – Inflation-Protected Fixed Income

**Members:** 10   **Var explained:** 0.906   **Signal:** LONG since 2026-05-27 (4 days, -0.7%)

#### Signal logic:

- Latest buy\_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell\_cross: 2026-05-18
- Current 252d return: +3.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-27 > last sell at 2026-05-18)



**Category breakdown**

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Inflation-Protected Fixed Income | 10        | 1          |

**Replication basket** – VTIP (cor=0.90) basket lead at lag 1: -0.011

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| VTIP   | 1.723  | Vanguard Short-Term Inflation-Protect... | Inflation-Protected F... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

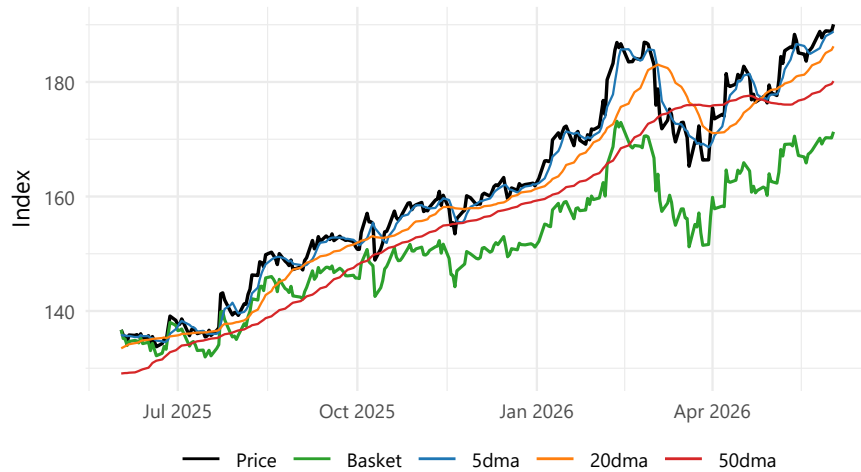
|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4  | TDTF   | 0.1034 | FlexShares iBoxx 5 Year Target Duration... | Inflation-Protected Fixe... | LONG       |
| 6  | TIPX   | 0.1033 | State Street SPDR Bloomberg 1a??10 Year... | Inflation-Protected Fixe... | FLAT       |
| 3  | TIP    | 0.1030 | iShares TIPS Bond ETF                      | Inflation-Protected Fixe... | LONG       |
| 8  | SCHP   | 0.1023 | Schwab U.S. TIPS ETF                       | Inflation-Protected Fixe... | LONG       |
| 5  | TDTT   | 0.1007 | FlexShares iBoxx 3 Year Target Duration... | Inflation-Protected Fixe... | FLAT       |
| 7  | SPIP   | 0.0999 | State Street SPDR Portfolio TIPS ETF       | Inflation-Protected Fixe... | LONG       |
| 9  | STIP   | 0.0998 | iShares 0-5 Year TIPS Bond ETF             | Inflation-Protected Fixe... | FLAT       |
| 2  | STPZ   | 0.0996 | PIMCO 1-5 Year U.S. TIPS Index Exchange... | Inflation-Protected Fixe... | FLAT       |
| 10 | VTIP   | 0.0994 | Vanguard Short-Term Inflation-Protected... | Inflation-Protected Fixe... | FLAT       |
| 1  | LTPZ   | 0.0886 | Pimco 15 Year U.S. TIPS Index Exchange...  | Inflation-Protected Fixe... | LONG       |

**Cluster 22 – Japan Equity**

**Members:** 9 **Var explained:** 0.887 **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell\_cross: 2026-04-29
- Current 252d return: +40.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Japan Equity        | 7         | 0.7799     |
| Asia-Pacific Equity | 2         | 0.2201     |

**Replication basket** – EWJ (cor=0.96) basket lead at lag 1: -0.037

| ticker | rep_wt | name                          | category     | etf_signal |
|--------|--------|-------------------------------|--------------|------------|
| EWJ    | 0.908  | iShares MSCI Japan Index Fund | Japan Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                   | category            | etf_signal |
|---|--------|--------|----------------------------------------|---------------------|------------|
| 1 | EWJ    | 0.1141 | iShares MSCI Japan Index Fund          | Japan Equity        | LONG       |
| 8 | BBJP   | 0.1141 | JPMorgan BetaBuilders Japan ETF        | Japan Equity        | LONG       |
| 9 | HEWJ   | 0.1116 | iShares Currency Hedged MSCI Japan ETF | Japan Equity        | LONG       |
| 6 | DBJP   | 0.1115 | Xtrackers MSCI Japan Hedged Equity ETF | Japan Equity        | LONG       |
| 3 | IPAC   | 0.1114 | iShares Core MSCI Pacific ETF          | Asia-Pacific Equity | FLAT       |
| 5 | FLJH   | 0.1107 | Franklin FTSE Japan Hedged ETF         | Japan Equity        | FLAT       |

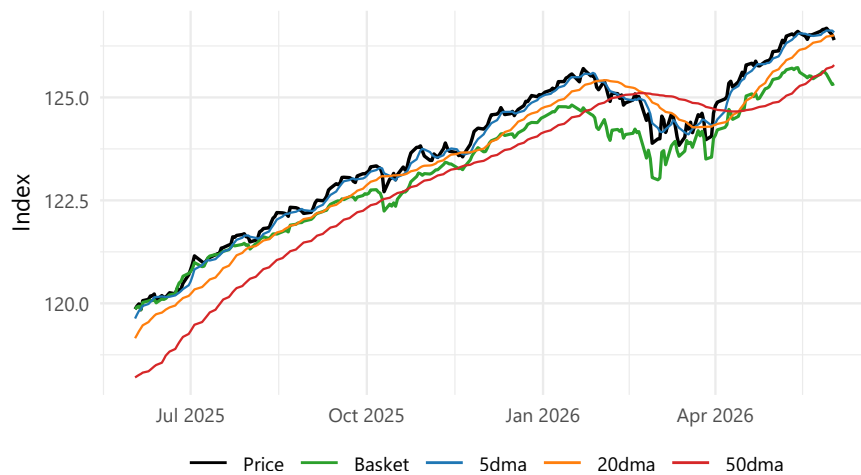
|   |      |        |                                     |                     |      |
|---|------|--------|-------------------------------------|---------------------|------|
| 2 | DXJ  | 0.1097 | WisdomTree Japan Hedged Equity Fund | Japan Equity        | LONG |
| 4 | VPL  | 0.1087 | Vanguard FTSE Pacific ETF           | Asia-Pacific Equity | LONG |
| 7 | EWJV | 0.1082 | iShares MSCI Japan Value ETF        | Japan Equity        | LONG |

### Cluster 23 – Bank Loan / Floating Rate

**Members:** 9 **Var explained:** 0.558 **Signal:** LONG since 2026-04-07 (39 days, 1.1%)

#### Signal logic:

- Latest buy\_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell\_cross: 2026-02-03
- Current 252d return: +5.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



#### Category breakdown

| category                          | n_members | abs_weight |
|-----------------------------------|-----------|------------|
| Bank Loan / Floating Rate         | 3         | 0.3628     |
| Interest Rate Hedged Fixed Income | 3         | 0.3244     |

|                            |   |        |
|----------------------------|---|--------|
| Floating Rate Fixed Income | 3 | 0.3128 |
|----------------------------|---|--------|

**Replication basket** – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| FLOT   | 0.293  | iShares Floating Rate Bond ETF           | Floating Rate Fixed I... | FLAT       |
| BKLN   | 0.211  | Invesco Senior Loan ETF                  | Bank Loan / Floating ... | LONG       |
| SRLN   | 0.308  | State Street Blackstone Senior Loan E... | Bank Loan / Floating ... | FLAT       |
| FLRN   | 0.126  | State Street SPDR Bloomberg Investmen... | Floating Rate Fixed I... | FLAT       |
| FLTR   | 0.035  | VanEck IG Floating Rate ETF              | Floating Rate Fixed I... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

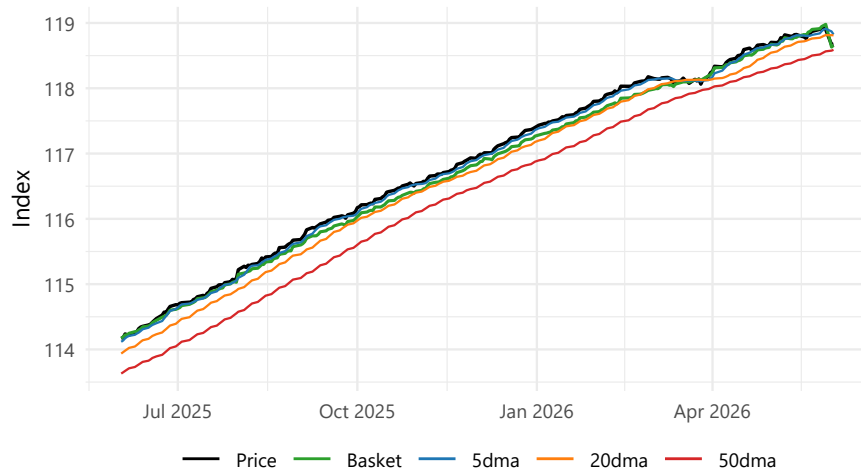
|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | SRLN   | 0.1250 | State Street Blackstone Senior Loan ETF    | Bank Loan / Floating Rate   | FLAT       |
| 3 | BKLN   | 0.1211 | Invesco Senior Loan ETF                    | Bank Loan / Floating Rate   | LONG       |
| 1 | FTSL   | 0.1167 | First Trust Senior Loan Fund               | Bank Loan / Floating Rate   | FLAT       |
| 5 | LQDH   | 0.1144 | iShares Interest Rate Hedged Corporate ... | Interest Rate Hedged Fix... | LONG       |
| 7 | FLOT   | 0.1083 | iShares Floating Rate Bond ETF             | Floating Rate Fixed Inco... | FLAT       |
| 6 | IGBH   | 0.1057 | iShares Interest Rate Hedged Long-Term ... | Interest Rate Hedged Fix... | LONG       |
| 4 | IGHG   | 0.1043 | ProShares Investment Grade-Interest Rat... | Interest Rate Hedged Fix... | LONG       |
| 8 | FLRN   | 0.1037 | State Street SPDR Bloomberg Investment ... | Floating Rate Fixed Inco... | FLAT       |
| 9 | FLTR   | 0.1009 | VanEck IG Floating Rate ETF                | Floating Rate Fixed Inco... | FLAT       |

## Cluster 24 – Cash / Ultra-Short Fixed Income

**Members:** 9 **Var explained:** 0.454 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell\_cross: 2026-03-26
- Current 252d return: +4.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-26 more recent than last effective buy 2022-11-29)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Cash / Ultra-Short Fixed Income | 9         | 1          |

**Replication basket** – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.097

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| SHV    | 0.298  | iShares 0-1 Year Treasury Bond ETF       | Cash / Ultra-Short Fi... | FLAT       |
| MINT   | 0.179  | PIMCO Enhanced Short Maturity Active ... | Cash / Ultra-Short Fi... | LONG       |
| GBIL   | 0.122  | Goldman Sachs Access Treasury 0-1 Yea... | Cash / Ultra-Short Fi... | LONG       |
| ICSH   | 0.406  | iShares Ultra Short Duration Bond Act... | Cash / Ultra-Short Fi... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 7 | SHV    | 0.1282 | iShares 0-1 Year Treasury Bond ETF         | Cash / Ultra-Short Fixed... | FLAT       |
| 6 | ICSH   | 0.1272 | iShares Ultra Short Duration Bond Activ... | Cash / Ultra-Short Fixed... | FLAT       |
| 5 | GSY    | 0.1263 | Invesco Ultra Short Duration ETF           | Cash / Ultra-Short Fixed... | FLAT       |
| 3 | FTSM   | 0.1201 | First Trust Enhanced Short Maturity ETF    | Cash / Ultra-Short Fixed... | FLAT       |

|   |      |        |                                            |                             |      |
|---|------|--------|--------------------------------------------|-----------------------------|------|
| 2 | MINT | 0.1100 | PIMCO Enhanced Short Maturity Active Ex... | Cash / Ultra-Short Fixed... | LONG |
| 4 | NEAR | 0.1096 | iShares Short Duration Bond Active ETF     | Cash / Ultra-Short Fixed... | FLAT |
| 1 | VNLA | 0.0987 | Janus Henderson Short Duration Income E... | Cash / Ultra-Short Fixed... | LONG |
| 8 | BIL  | 0.0954 | State Street SPDR Bloomberg 1-3 Month T... | Cash / Ultra-Short Fixed... | LONG |
| 9 | GBIL | 0.0844 | Goldman Sachs Access Treasury 0-1 Year ... | Cash / Ultra-Short Fixed... | LONG |

## Cluster 25 – International Fixed Income

**Members:** 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (2 days, -0.5%)

### Signal logic:

- Latest buy\_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell\_cross: 2026-05-18
- Current 252d return: -0.4% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
| Currency | 4         | 0.641      |

|                            |   |       |
|----------------------------|---|-------|
| International Fixed Income | 2 | 0.359 |
|----------------------------|---|-------|

**Replication basket** – BWX (cor=0.93) basket lead at lag 1: 0.027

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| BWX    | 0.751  | State Street SPDR Bloomberg Internati... | International Fixed I... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | IGOV   | 0.1801 | iShares International Treasury Bond ETF    | International Fixed Inco... | LONG       |
| 1 | BWX    | 0.1790 | State Street SPDR Bloomberg Internation... | International Fixed Inco... | FLAT       |
| 4 | FXE    | 0.1701 | Invesco CurrencyShares Euro Currency Tr... | Currency                    | FLAT       |
| 5 | FXF    | 0.1648 | Invesco CurrencyShares Swiss Franc Trust   | Currency                    | FLAT       |
| 3 | FXB    | 0.1610 | Invesco CurrencyShares British Pound St... | Currency                    | LONG       |
| 6 | FXY    | 0.1451 | Invesco CurrencyShares Japanese Yen Tru... | Currency                    | FLAT       |

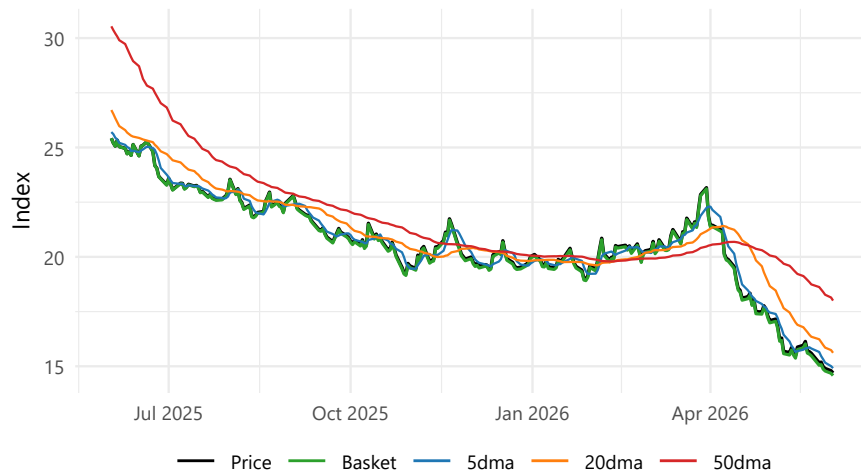
## Cluster 26 – Leveraged / Inverse

**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell\_cross: 2026-04-08
- Current 252d return: -43.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)





### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Currency            | 1         | 0.5        |
| Leveraged / Inverse | 1         | 0.5        |

**Replication basket** – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.018

| ticker | rep_wt | name                                  | category            | etf_signal |
|--------|--------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.507  | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.495  | Invesco DB USD Index Bullish Fund ETF | Currency            | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt | name                                  | category            | etf_signal |
|--------|-------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.5   | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.5   | Invesco DB USD Index Bullish Fund ETF | Currency            | LONG       |

**Long ETF Positions (full)**

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

| ticker | cluster   | theme                 | entry      | days | return | p_sell | nss  |
|--------|-----------|-----------------------|------------|------|--------|--------|------|
| QQEW   | cluster_1 | U.S. Large Cap Equity | 2026-04-16 | 32   | 18.9%  | 0.00   | 1.00 |
| LRGF   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 11.1%  | 0.00   | 1.00 |
| FTEC   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 35.6%  | 0.00   | 1.00 |
| VGT    | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 35.4%  | 0.00   | 1.00 |
| TDIV   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 29.3%  | 0.00   | 1.00 |
| NXTG   | cluster_1 | U.S. Large Cap Equity | 2026-04-08 | 38   | 39.3%  | 0.00   | 1.00 |
| XNTK   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 40.1%  | 0.00   | 1.00 |
| XLK    | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 39.0%  | 0.00   | 1.00 |
| IGM    | cluster_1 | U.S. Large Cap Equity | 2026-04-09 | 37   | 34.5%  | 0.00   | 1.00 |
| DGRW   | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34   | 6.4%   | 0.00   | 1.00 |
| IXN    | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 38.8%  | 0.00   | 1.00 |
| VONG   | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34   | 11.4%  | 0.00   | 1.00 |
| FXL    | cluster_1 | U.S. Large Cap Equity | 2026-04-09 | 37   | 33.8%  | 0.00   | 1.00 |
| IWF    | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34   | 11.4%  | 0.00   | 1.00 |
| QTEC   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 47.1%  | 0.00   | 1.00 |
| QTUM   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 44.6%  | 0.00   | 1.00 |
| GSLC   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 9.4%   | 0.00   | 1.00 |
| AIQ    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 38.7%  | 0.00   | 1.00 |
| SUSA   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 12.3%  | 0.00   | 1.00 |
| EPS    | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 11.6%  | 0.00   | 1.00 |
| SPHQ   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 9.3%   | 0.00   | 1.00 |
| IWY    | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34   | 11.9%  | 0.00   | 1.00 |
| ESGU   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 11.7%  | 0.00   | 1.00 |
| QUAL   | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34   | 6.2%   | 0.00   | 1.00 |
| QDF    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 8.6%   | 0.00   | 1.00 |
| NULG   | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 20.1%  | 0.00   | 1.00 |
| VONE   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.3%  | 0.00   | 1.00 |
| MGK    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 16.6%  | 0.00   | 1.00 |
| IWB    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.3%  | 0.00   | 1.00 |
| VV     | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 11.0%  | 0.00   | 1.00 |
| SCHX   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.7%  | 0.00   | 1.00 |
| ILCB   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 11.0%  | 0.00   | 1.00 |
| JHML   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 8.8%   | 0.00   | 1.00 |
| VTI    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.5%  | 0.00   | 1.00 |
| IWV    | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.3%  | 0.00   | 1.00 |
| IYW    | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36   | 33.8%  | 0.00   | 1.00 |
| SCHB   | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35   | 10.6%  | 0.00   | 1.00 |

|      |           |                       |            |    |       |      |      |
|------|-----------|-----------------------|------------|----|-------|------|------|
| IYY  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.6% | 0.00 | 1.00 |
| ITOT | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.6% | 0.00 | 1.00 |
| VTHR | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.4% | 0.00 | 1.00 |
| IWL  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 11.4% | 0.00 | 1.00 |
| VUG  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 15.6% | 0.00 | 1.00 |
| SPYX | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.8% | 0.00 | 1.00 |
| VOO  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.7% | 0.00 | 1.00 |
| MGC  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 11.9% | 0.00 | 1.00 |
| IVV  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.7% | 0.00 | 1.00 |
| SPY  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.7% | 0.00 | 1.00 |
| PTLC | cluster_1 | U.S. Large Cap Equity | 2026-04-20 | 30 | 7.1%  | 0.00 | 1.00 |
| SPTM | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 10.2% | 0.00 | 1.00 |
| ILCG | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 16.5% | 0.00 | 1.00 |
| PSCT | cluster_1 | U.S. Large Cap Equity | 2026-04-07 | 39 | 42.2% | 0.00 | 1.00 |
| TILT | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 8.9%  | 0.00 | 1.00 |
| SCHG | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 13.4% | 0.00 | 1.00 |
| QUS  | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34 | 4.3%  | 0.00 | 1.00 |
| OMFL | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36 | 9.3%  | 0.00 | 1.00 |
| QQQ  | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36 | 22.1% | 0.00 | 1.00 |
| PWB  | cluster_1 | U.S. Large Cap Equity | 2026-04-10 | 36 | 19.5% | 0.00 | 1.00 |
| PTNQ | cluster_1 | U.S. Large Cap Equity | 2026-04-17 | 31 | 15.0% | 0.00 | 1.00 |
| SOXX | cluster_1 | U.S. Large Cap Equity | 2026-04-08 | 38 | 63.3% | 0.00 | 1.00 |
| CATH | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 9.8%  | 0.00 | 1.00 |
| IVW  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 15.7% | 0.00 | 1.00 |
| VOOG | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 15.7% | 0.00 | 1.00 |
| SPYG | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 15.8% | 0.00 | 1.00 |
| OEF  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 12.5% | 0.00 | 1.00 |
| IUSG | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 15.3% | 0.00 | 1.00 |
| ONEQ | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 16.9% | 0.00 | 1.00 |
| XSD  | cluster_1 | U.S. Large Cap Equity | 2026-04-09 | 37 | 75.7% | 0.00 | 1.00 |
| SMH  | cluster_1 | U.S. Large Cap Equity | 2026-04-08 | 38 | 49.5% | 0.00 | 1.00 |
| PTF  | cluster_1 | U.S. Large Cap Equity | 2026-04-07 | 39 | 45.6% | 0.00 | 1.00 |
| IOO  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 12.2% | 0.00 | 1.00 |
| FTXL | cluster_1 | U.S. Large Cap Equity | 2026-04-07 | 39 | 74.7% | 0.00 | 1.00 |
| XLG  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 12.1% | 0.00 | 1.00 |
| DSI  | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34 | 11.2% | 0.01 | 0.99 |
| PSI  | cluster_1 | U.S. Large Cap Equity | 2026-04-07 | 39 | 62.4% | 0.06 | 0.94 |
| IYC  | cluster_1 | U.S. Large Cap Equity | 2026-04-14 | 34 | -1.6% | 0.09 | 0.90 |
| VOX  | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 1.1%  | 0.63 | 0.37 |
| FCOM | cluster_1 | U.S. Large Cap Equity | 2026-04-13 | 35 | 0.9%  | 0.64 | 0.36 |
| CIBR | cluster_5 | Global Equity         | 2026-04-17 | 31 | 41.9% | 0.00 | 1.00 |
| HACK | cluster_5 | Global Equity         | 2026-04-17 | 31 | 34.1% | 0.00 | 1.00 |
| IGV  | cluster_5 | Global Equity         | 2026-04-21 | 29 | 20.8% | 0.00 | 1.00 |
| PRNT | cluster_5 | Global Equity         | 2026-04-16 | 32 | 16.3% | 0.00 | 1.00 |
| IPO  | cluster_5 | Global Equity         | 2026-04-13 | 35 | 24.6% | 0.00 | 1.00 |

|      |            |               |            |    |       |      |      |
|------|------------|---------------|------------|----|-------|------|------|
| XT   | cluster_5  | Global Equity | 2026-04-13 | 35 | 17.3% | 0.00 | 1.00 |
| FDN  | cluster_5  | Global Equity | 2026-04-10 | 36 | 17.7% | 0.00 | 1.00 |
| IWP  | cluster_5  | Global Equity | 2026-04-15 | 33 | 6.1%  | 0.00 | 1.00 |
| FPXI | cluster_5  | Global Equity | 2026-04-10 | 36 | 16.9% | 0.00 | 1.00 |
| SKYY | cluster_5  | Global Equity | 2026-04-17 | 31 | 30.7% | 0.00 | 1.00 |
| IMCG | cluster_5  | Global Equity | 2026-04-09 | 37 | 15.3% | 0.00 | 1.00 |
| URTH | cluster_5  | Global Equity | 2026-04-13 | 35 | 8.5%  | 0.00 | 1.00 |
| SPHB | cluster_5  | Global Equity | 2026-04-10 | 36 | 25.3% | 0.00 | 1.00 |
| CRBN | cluster_5  | Global Equity | 2026-04-13 | 35 | 9.3%  | 0.00 | 1.00 |
| EWS  | cluster_5  | Global Equity | 2026-04-07 | 39 | 5.6%  | 0.00 | 1.00 |
| VOT  | cluster_5  | Global Equity | 2026-04-14 | 34 | 10.0% | 0.00 | 1.00 |
| ACWI | cluster_5  | Global Equity | 2026-04-13 | 35 | 9.3%  | 0.00 | 1.00 |
| VT   | cluster_5  | Global Equity | 2026-04-13 | 35 | 8.8%  | 0.00 | 1.00 |
| RPG  | cluster_5  | Global Equity | 2026-04-09 | 37 | 20.1% | 0.00 | 1.00 |
| ARKQ | cluster_5  | Global Equity | 2026-04-14 | 34 | 14.8% | 0.00 | 1.00 |
| FTC  | cluster_5  | Global Equity | 2026-04-10 | 36 | 14.2% | 0.00 | 1.00 |
| AOA  | cluster_5  | Global Equity | 2026-04-13 | 35 | 6.9%  | 0.00 | 1.00 |
| VBK  | cluster_5  | Global Equity | 2026-04-10 | 36 | 13.4% | 0.00 | 1.00 |
| ICVT | cluster_5  | Global Equity | 2026-04-08 | 38 | 15.5% | 0.00 | 1.00 |
| CWB  | cluster_5  | Global Equity | 2026-04-08 | 38 | 15.0% | 0.00 | 1.00 |
| MTUM | cluster_5  | Global Equity | 2026-04-08 | 38 | 25.6% | 0.00 | 1.00 |
| AOR  | cluster_5  | Global Equity | 2026-04-13 | 35 | 5.1%  | 0.00 | 1.00 |
| SPGM | cluster_5  | Global Equity | 2026-04-10 | 36 | 9.1%  | 0.00 | 1.00 |
| AOM  | cluster_5  | Global Equity | 2026-04-13 | 35 | 3.5%  | 0.00 | 1.00 |
| BLOK | cluster_5  | Global Equity | 2026-04-13 | 35 | 21.9% | 0.00 | 1.00 |
| ARKW | cluster_5  | Global Equity | 2026-04-14 | 34 | 15.4% | 0.00 | 1.00 |
| AOK  | cluster_5  | Global Equity | 2026-04-13 | 35 | 2.5%  | 0.00 | 1.00 |
| FPX  | cluster_5  | Global Equity | 2026-04-08 | 38 | 15.0% | 0.01 | 0.99 |
| QAI  | cluster_5  | Global Equity | 2026-04-08 | 38 | 5.3%  | 0.01 | 0.99 |
| IWO  | cluster_5  | Global Equity | 2026-04-13 | 35 | 12.8% | 0.01 | 0.99 |
| VTWG | cluster_5  | Global Equity | 2026-04-13 | 35 | 13.0% | 0.01 | 0.99 |
| ROBO | cluster_5  | Global Equity | 2026-04-14 | 34 | 18.5% | 0.01 | 0.99 |
| IBUY | cluster_5  | Global Equity | 2026-05-27 | 4  | 1.1%  | 0.01 | 0.93 |
| BLCN | cluster_5  | Global Equity | 2026-04-15 | 33 | 15.2% | 0.07 | 0.93 |
| PDP  | cluster_5  | Global Equity | 2026-04-08 | 38 | 11.8% | 0.08 | 0.92 |
| BOTZ | cluster_5  | Global Equity | 2026-04-16 | 32 | 10.7% | 0.64 | 0.36 |
| RSPH | cluster_11 | Healthcare    | 2026-05-21 | 7  | -0.4% | 0.00 | 1.00 |
| IYH  | cluster_11 | Healthcare    | 2026-05-20 | 8  | -0.5% | 0.00 | 1.00 |
| FHLC | cluster_11 | Healthcare    | 2026-05-19 | 9  | -0.4% | 0.00 | 1.00 |
| VHT  | cluster_11 | Healthcare    | 2026-05-19 | 9  | -0.4% | 0.00 | 1.00 |
| XLV  | cluster_11 | Healthcare    | 2026-05-21 | 7  | -1.2% | 0.00 | 1.00 |
| PPH  | cluster_11 | Healthcare    | 2026-05-21 | 7  | -2.5% | 0.01 | 0.99 |
| IXJ  | cluster_11 | Healthcare    | 2026-05-22 | 6  | -3.1% | 0.01 | 0.98 |
| FXH  | cluster_11 | Healthcare    | 2026-04-10 | 36 | 0.8%  | 0.03 | 0.97 |
| IBB  | cluster_11 | Healthcare    | 2026-05-28 | 3  | -4.6% | 0.23 | 0.40 |

|      |            |                      |            |    |       |      |       |
|------|------------|----------------------|------------|----|-------|------|-------|
| IHF  | cluster_11 | Healthcare           | 2026-05-08 | 16 | -0.1% | 0.80 | 0.20  |
| PSCH | cluster_11 | Healthcare           | 2026-04-13 | 35 | 3.3%  | 0.94 | 0.06  |
| XHE  | cluster_11 | Healthcare           | 2026-05-22 | 6  | -3.8% | 0.46 | -0.26 |
| FDVV | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 6.6%  | 0.00 | 1.00  |
| VIG  | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 5.4%  | 0.00 | 1.00  |
| PFM  | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 5.0%  | 0.00 | 1.00  |
| RPV  | cluster_3  | U.S. Large Cap Value | 2026-04-20 | 30 | 3.8%  | 0.00 | 1.00  |
| DIA  | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 5.9%  | 0.00 | 1.00  |
| SDOG | cluster_3  | U.S. Large Cap Value | 2026-04-07 | 39 | 6.2%  | 0.00 | 1.00  |
| RWL  | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 7.5%  | 0.00 | 1.00  |
| JVAL | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 15.3% | 0.00 | 1.00  |
| VLUE | cluster_3  | U.S. Large Cap Value | 2026-04-09 | 37 | 33.3% | 0.00 | 1.00  |
| DGRO | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 4.2%  | 0.00 | 1.00  |
| RSP  | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 5.2%  | 0.00 | 1.00  |
| SCHV | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 7.9%  | 0.00 | 1.00  |
| USMV | cluster_3  | U.S. Large Cap Value | 2026-05-05 | 19 | 2.6%  | 0.00 | 1.00  |
| MGV  | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 6.1%  | 0.00 | 1.00  |
| IMCB | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 9.6%  | 0.00 | 1.00  |
| SIZE | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 6.4%  | 0.00 | 1.00  |
| VTV  | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 5.3%  | 0.00 | 1.00  |
| DLN  | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 5.2%  | 0.00 | 1.00  |
| VYM  | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 6.0%  | 0.00 | 1.00  |
| DTD  | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 4.9%  | 0.00 | 1.00  |
| ILCV | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 5.8%  | 0.00 | 1.00  |
| FEX  | cluster_3  | U.S. Large Cap Value | 2026-04-09 | 37 | 7.7%  | 0.00 | 1.00  |
| FNDX | cluster_3  | U.S. Large Cap Value | 2026-04-09 | 37 | 8.0%  | 0.00 | 1.00  |
| RDIV | cluster_3  | U.S. Large Cap Value | 2026-04-21 | 29 | 4.3%  | 0.00 | 1.00  |
| AIVL | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 5.3%  | 0.00 | 1.00  |
| ONEY | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 5.4%  | 0.00 | 1.00  |
| PRF  | cluster_3  | U.S. Large Cap Value | 2026-04-09 | 37 | 8.7%  | 0.00 | 1.00  |
| VO   | cluster_3  | U.S. Large Cap Value | 2026-04-13 | 35 | 6.6%  | 0.00 | 1.00  |
| IWR  | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 7.5%  | 0.00 | 1.00  |
| VOOV | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 4.8%  | 0.00 | 1.00  |
| FTA  | cluster_3  | U.S. Large Cap Value | 2026-05-21 | 7  | 1.9%  | 0.00 | 1.00  |
| IVE  | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 4.7%  | 0.00 | 1.00  |
| SPYV | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 4.7%  | 0.00 | 1.00  |
| IUSV | cluster_3  | U.S. Large Cap Value | 2026-04-14 | 34 | 4.6%  | 0.00 | 1.00  |
| OUSA | cluster_3  | U.S. Large Cap Value | 2026-04-20 | 30 | 1.0%  | 0.00 | 1.00  |
| IWD  | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 7.9%  | 0.00 | 1.00  |
| VONV | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 7.9%  | 0.00 | 1.00  |
| IWX  | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 8.4%  | 0.00 | 1.00  |
| ACWV | cluster_3  | U.S. Large Cap Value | 2026-05-06 | 18 | 1.0%  | 0.01 | 0.99  |
| SPYD | cluster_3  | U.S. Large Cap Value | 2026-04-21 | 29 | 2.7%  | 0.01 | 0.99  |
| NULV | cluster_3  | U.S. Large Cap Value | 2026-04-10 | 36 | 9.1%  | 0.02 | 0.98  |
| SCHD | cluster_3  | U.S. Large Cap Value | 2026-04-21 | 29 | 4.5%  | 0.02 | 0.98  |

|      |           |                        |            |    |       |      |       |
|------|-----------|------------------------|------------|----|-------|------|-------|
| PWV  | cluster_3 | U.S. Large Cap Value   | 2026-04-02 | 41 | 6.4%  | 0.07 | 0.93  |
| SDY  | cluster_3 | U.S. Large Cap Value   | 2026-05-20 | 8  | 0.5%  | 0.11 | 0.89  |
| PKW  | cluster_3 | U.S. Large Cap Value   | 2026-05-20 | 8  | 1.3%  | 0.10 | 0.80  |
| DHS  | cluster_3 | U.S. Large Cap Value   | 2026-04-20 | 30 | 0.8%  | 0.47 | 0.53  |
| FDL  | cluster_3 | U.S. Large Cap Value   | 2026-05-19 | 9  | -1.0% | 0.35 | 0.23  |
| FTCS | cluster_3 | U.S. Large Cap Value   | 2026-05-20 | 8  | -1.6% | 0.42 | 0.21  |
| SPHD | cluster_3 | U.S. Large Cap Value   | 2026-05-18 | 10 | -0.3% | 0.75 | 0.03  |
| HDV  | cluster_3 | U.S. Large Cap Value   | 2026-05-01 | 21 | -0.7% | 0.85 | 0.01  |
| FVD  | cluster_3 | U.S. Large Cap Value   | 2026-05-22 | 6  | -1.6% | 0.37 | -0.02 |
| FIW  | cluster_3 | U.S. Large Cap Value   | 2026-05-08 | 16 | -1.4% | 0.25 | -0.11 |
| LVHD | cluster_3 | U.S. Large Cap Value   | 2026-05-22 | 6  | -2.4% | 0.76 | -0.61 |
| FTXR | cluster_2 | U.S. Mid Cap Equity    | 2026-04-17 | 31 | 6.9%  | 0.00 | 1.00  |
| FV   | cluster_2 | U.S. Mid Cap Equity    | 2026-04-14 | 34 | 14.2% | 0.00 | 1.00  |
| XAR  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-22 | 6  | 3.0%  | 0.00 | 1.00  |
| PPA  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-27 | 4  | -0.8% | 0.00 | 1.00  |
| PEJ  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-26 | 5  | 2.0%  | 0.00 | 1.00  |
| ISMD | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 12.5% | 0.00 | 1.00  |
| ITA  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-26 | 5  | -0.5% | 0.00 | 1.00  |
| SCHM | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 10.0% | 0.01 | 0.99  |
| SCHA | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 11.2% | 0.01 | 0.99  |
| FXD  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-27 | 4  | -0.8% | 0.00 | 0.99  |
| IVOG | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 8.9%  | 0.02 | 0.98  |
| IJS  | cluster_2 | U.S. Mid Cap Equity    | 2026-04-10 | 36 | 7.7%  | 0.02 | 0.98  |
| MDYG | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 8.9%  | 0.02 | 0.98  |
| IJK  | cluster_2 | U.S. Mid Cap Equity    | 2026-04-09 | 37 | 8.9%  | 0.02 | 0.98  |
| PSCD | cluster_2 | U.S. Mid Cap Equity    | 2026-05-27 | 4  | 0.4%  | 0.01 | 0.95  |
| VFMO | cluster_2 | U.S. Mid Cap Equity    | 2026-04-08 | 38 | 12.5% | 0.05 | 0.95  |
| RSPD | cluster_2 | U.S. Mid Cap Equity    | 2026-05-28 | 3  | -1.4% | 0.00 | 0.93  |
| FXR  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-26 | 5  | 0.2%  | 0.06 | 0.92  |
| XMMO | cluster_2 | U.S. Mid Cap Equity    | 2026-04-06 | 40 | 15.2% | 0.09 | 0.91  |
| XRT  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-28 | 3  | -2.2% | 0.04 | 0.79  |
| XHB  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-29 | 2  | 1.6%  | 0.05 | 0.72  |
| ITB  | cluster_2 | U.S. Mid Cap Equity    | 2026-06-01 | 1  | 0.2%  | 0.06 | 0.57  |
| PKB  | cluster_2 | U.S. Mid Cap Equity    | 2026-05-28 | 3  | -1.2% | 0.44 | 0.13  |
| IHDG | cluster_4 | Developed ex-US Equity | 2026-05-06 | 18 | 1.2%  | 0.00 | 1.00  |
| VIDI | cluster_4 | Developed ex-US Equity | 2026-04-09 | 37 | 10.1% | 0.00 | 1.00  |
| EWO  | cluster_4 | Developed ex-US Equity | 2026-04-10 | 36 | 7.6%  | 0.00 | 1.00  |
| HFXI | cluster_4 | Developed ex-US Equity | 2026-04-09 | 37 | 7.9%  | 0.00 | 1.00  |
| FNDF | cluster_4 | Developed ex-US Equity | 2026-04-08 | 38 | 7.4%  | 0.00 | 1.00  |
| PXF  | cluster_4 | Developed ex-US Equity | 2026-04-08 | 38 | 7.5%  | 0.00 | 1.00  |
| EWN  | cluster_4 | Developed ex-US Equity | 2026-04-10 | 36 | 9.3%  | 0.00 | 1.00  |
| SCHF | cluster_4 | Developed ex-US Equity | 2026-04-10 | 36 | 7.2%  | 0.00 | 1.00  |
| EWD  | cluster_4 | Developed ex-US Equity | 2026-05-21 | 7  | 1.0%  | 0.00 | 1.00  |
| SPDW | cluster_4 | Developed ex-US Equity | 2026-04-10 | 36 | 6.9%  | 0.00 | 1.00  |
| EWG  | cluster_4 | Developed ex-US Equity | 2026-04-14 | 34 | 3.3%  | 0.00 | 1.00  |

|      |            |                         |            |    |       |      |      |
|------|------------|-------------------------|------------|----|-------|------|------|
| ACWX | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 7.3%  | 0.00 | 1.00 |
| VEA  | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 6.8%  | 0.00 | 1.00 |
| VEU  | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 7.3%  | 0.01 | 0.99 |
| IXUS | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 7.2%  | 0.01 | 0.99 |
| VXUS | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 7.0%  | 0.01 | 0.99 |
| VIGI | cluster_4  | Developed ex-US Equity  | 2026-04-14 | 34 | 1.9%  | 0.01 | 0.99 |
| CWI  | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 6.1%  | 0.01 | 0.99 |
| IQDF | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 6.7%  | 0.02 | 0.98 |
| FEP  | cluster_4  | Developed ex-US Equity  | 2026-04-09 | 37 | 3.3%  | 0.05 | 0.95 |
| VYMI | cluster_4  | Developed ex-US Equity  | 2026-04-09 | 37 | 2.3%  | 0.08 | 0.92 |
| SCZ  | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 3.3%  | 0.09 | 0.91 |
| EWI  | cluster_4  | Developed ex-US Equity  | 2026-04-08 | 38 | 5.1%  | 0.10 | 0.90 |
| IMTM | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 3.6%  | 0.15 | 0.85 |
| DLS  | cluster_4  | Developed ex-US Equity  | 2026-04-13 | 35 | 0.6%  | 0.38 | 0.62 |
| FNDC | cluster_4  | Developed ex-US Equity  | 2026-04-10 | 36 | 2.3%  | 0.39 | 0.61 |
| IPKW | cluster_4  | Developed ex-US Equity  | 2026-04-09 | 37 | 2.0%  | 0.67 | 0.31 |
| IDV  | cluster_4  | Developed ex-US Equity  | 2026-04-06 | 40 | 3.5%  | 0.79 | 0.20 |
| EEMV | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 12.4% | 0.00 | 1.00 |
| EWT  | cluster_7  | Emerging Markets Equity | 2026-04-08 | 38 | 40.3% | 0.00 | 1.00 |
| EWY  | cluster_7  | Emerging Markets Equity | 2026-04-09 | 37 | 54.0% | 0.00 | 1.00 |
| AIA  | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 31.8% | 0.00 | 1.00 |
| DEM  | cluster_7  | Emerging Markets Equity | 2026-04-09 | 37 | 10.3% | 0.00 | 1.00 |
| XSOE | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 18.1% | 0.00 | 1.00 |
| GEM  | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 16.1% | 0.00 | 1.00 |
| ESGE | cluster_7  | Emerging Markets Equity | 2026-04-13 | 35 | 16.2% | 0.00 | 1.00 |
| AAXJ | cluster_7  | Emerging Markets Equity | 2026-04-13 | 35 | 19.4% | 0.00 | 1.00 |
| EEM  | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 16.9% | 0.00 | 1.00 |
| EMGF | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 17.9% | 0.01 | 0.99 |
| EEMA | cluster_7  | Emerging Markets Equity | 2026-04-13 | 35 | 18.4% | 0.01 | 0.99 |
| IEMG | cluster_7  | Emerging Markets Equity | 2026-04-10 | 36 | 15.8% | 0.01 | 0.99 |
| SPEM | cluster_7  | Emerging Markets Equity | 2026-04-13 | 35 | 7.4%  | 0.06 | 0.94 |
| FEMS | cluster_7  | Emerging Markets Equity | 2026-05-28 | 3  | 1.4%  | 0.56 | 0.00 |
| DBJP | cluster_22 | Japan Equity            | 2026-04-08 | 38 | 5.5%  | 0.00 | 1.00 |
| HEWJ | cluster_22 | Japan Equity            | 2026-04-08 | 38 | 5.9%  | 0.00 | 1.00 |
| VPL  | cluster_22 | Japan Equity            | 2026-04-10 | 36 | 13.6% | 0.01 | 0.99 |
| EWJ  | cluster_22 | Japan Equity            | 2026-04-13 | 35 | 5.9%  | 0.06 | 0.94 |
| DXJ  | cluster_22 | Japan Equity            | 2026-05-01 | 21 | 6.1%  | 0.06 | 0.94 |
| EWJV | cluster_22 | Japan Equity            | 2026-05-06 | 18 | 2.3%  | 0.07 | 0.93 |
| BBJP | cluster_22 | Japan Equity            | 2026-04-30 | 22 | 4.5%  | 0.08 | 0.92 |
| SHM  | cluster_16 | Municipal Bond          | 2026-05-28 | 3  | -0.1% | 0.00 | 1.00 |
| SUB  | cluster_16 | Municipal Bond          | 2026-05-27 | 4  | -0.0% | 0.00 | 0.99 |
| HYD  | cluster_16 | Municipal Bond          | 2026-05-22 | 6  | 1.1%  | 0.01 | 0.99 |
| ITM  | cluster_16 | Municipal Bond          | 2026-05-28 | 3  | 0.1%  | 0.01 | 0.99 |
| VTEB | cluster_16 | Municipal Bond          | 2026-05-28 | 3  | 0.1%  | 0.01 | 0.98 |
| HYMB | cluster_16 | Municipal Bond          | 2026-05-26 | 5  | 0.4%  | 0.02 | 0.98 |

|      |            |                        |            |    |       |      |       |
|------|------------|------------------------|------------|----|-------|------|-------|
| TFI  | cluster_16 | Municipal Bond         | 2026-05-28 | 3  | 0.1%  | 0.02 | 0.97  |
| CMF  | cluster_16 | Municipal Bond         | 2026-05-28 | 3  | 0.1%  | 0.02 | 0.97  |
| MUB  | cluster_16 | Municipal Bond         | 2026-05-28 | 3  | 0.1%  | 0.03 | 0.96  |
| NYF  | cluster_16 | Municipal Bond         | 2026-05-28 | 3  | 0.2%  | 0.04 | 0.94  |
| MUNI | cluster_16 | Municipal Bond         | 2026-05-28 | 3  | -0.1% | 0.05 | 0.92  |
| FMB  | cluster_16 | Municipal Bond         | 2026-05-29 | 2  | 0.3%  | 0.08 | 0.89  |
| PZA  | cluster_16 | Municipal Bond         | 2026-05-29 | 2  | 0.5%  | 0.07 | 0.86  |
| SPLB | cluster_13 | Fixed Income           | 2026-05-27 | 4  | 0.2%  | 0.00 | 0.99  |
| VCLT | cluster_13 | Fixed Income           | 2026-05-27 | 4  | 0.1%  | 0.00 | 0.99  |
| IGLB | cluster_13 | Fixed Income           | 2026-05-27 | 4  | 0.1%  | 0.00 | 0.99  |
| USIG | cluster_13 | Fixed Income           | 2026-05-27 | 4  | -0.1% | 0.02 | 0.96  |
| LQD  | cluster_13 | Fixed Income           | 2026-05-28 | 3  | -0.3% | 0.01 | 0.96  |
| FLCO | cluster_13 | Fixed Income           | 2026-05-27 | 4  | -0.2% | 0.02 | 0.95  |
| CORP | cluster_13 | Fixed Income           | 2026-05-28 | 3  | -0.3% | 0.03 | 0.92  |
| QLTA | cluster_13 | Fixed Income           | 2026-05-28 | 3  | -0.4% | 0.02 | 0.92  |
| BLV  | cluster_13 | Fixed Income           | 2026-05-29 | 2  | -0.3% | 0.02 | 0.88  |
| EDV  | cluster_13 | Fixed Income           | 2026-06-02 | 0  | 0.0%  | 0.01 | 0.75  |
| VWOB | cluster_10 | High Yield Bond        | 2026-05-26 | 5  | 0.6%  | 0.02 | 0.98  |
| HYEM | cluster_10 | High Yield Bond        | 2026-04-13 | 35 | 1.2%  | 0.04 | 0.96  |
| PCY  | cluster_10 | High Yield Bond        | 2026-05-28 | 3  | 0.7%  | 0.06 | 0.91  |
| FPE  | cluster_10 | High Yield Bond        | 2026-05-27 | 4  | 0.1%  | 0.90 | -0.27 |
| SLX  | cluster_9  | Materials              | 2026-04-09 | 37 | 15.8% | 0.01 | 0.99  |
| XME  | cluster_9  | Materials              | 2026-05-26 | 5  | 8.3%  | 0.01 | 0.99  |
| COPX | cluster_9  | Materials              | 2026-05-08 | 16 | 8.9%  | 0.02 | 0.98  |
| DBB  | cluster_9  | Materials              | 2026-04-10 | 36 | 9.7%  | 0.03 | 0.97  |
| IYM  | cluster_9  | Materials              | 2026-05-26 | 5  | 1.2%  | 0.12 | 0.85  |
| VAW  | cluster_9  | Materials              | 2026-05-28 | 3  | 0.5%  | 0.16 | 0.65  |
| XLB  | cluster_9  | Materials              | 2026-05-28 | 3  | 0.3%  | 0.31 | 0.34  |
| RSPM | cluster_9  | Materials              | 2026-05-29 | 2  | 0.5%  | 0.31 | 0.22  |
| IAGG | cluster_6  | Aggregate Fixed Income | 2026-05-22 | 6  | 0.5%  | 0.00 | 1.00  |
| BNDX | cluster_6  | Aggregate Fixed Income | 2026-05-26 | 5  | -0.1% | 0.00 | 1.00  |
| SPIB | cluster_6  | Aggregate Fixed Income | 2026-05-27 | 4  | -0.3% | 0.06 | 0.83  |
| IGSB | cluster_6  | Aggregate Fixed Income | 2026-05-26 | 5  | -0.2% | 0.12 | 0.82  |
| IGIB | cluster_6  | Aggregate Fixed Income | 2026-05-28 | 3  | -0.4% | 0.06 | 0.82  |
| VCSH | cluster_6  | Aggregate Fixed Income | 2026-05-26 | 5  | -0.2% | 0.14 | 0.78  |
| VCIT | cluster_6  | Aggregate Fixed Income | 2026-05-28 | 3  | -0.4% | 0.07 | 0.78  |
| IUSB | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.3% | 0.06 | 0.77  |
| VMBS | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.6% | 0.08 | 0.68  |
| AGGY | cluster_6  | Aggregate Fixed Income | 2026-06-01 | 1  | 0.0%  | 0.08 | 0.68  |
| AGG  | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.4% | 0.07 | 0.68  |
| SCHZ | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.4% | 0.06 | 0.67  |
| BND  | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.4% | 0.07 | 0.67  |
| MBB  | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.5% | 0.07 | 0.67  |
| SPAB | cluster_6  | Aggregate Fixed Income | 2026-05-29 | 2  | -0.4% | 0.07 | 0.65  |
| ISTB | cluster_6  | Aggregate Fixed Income | 2026-05-27 | 4  | -0.2% | 0.16 | 0.64  |



|      |            |                            |            |      |       |      |       |
|------|------------|----------------------------|------------|------|-------|------|-------|
| BOND | cluster_6  | Aggregate Fixed Income     | 2026-05-29 | 2    | -0.4% | 0.09 | 0.63  |
| SPMB | cluster_6  | Aggregate Fixed Income     | 2026-05-29 | 2    | -0.4% | 0.07 | 0.61  |
| GTO  | cluster_6  | Aggregate Fixed Income     | 2026-06-01 | 1    | 0.0%  | 0.13 | 0.57  |
| GNMA | cluster_6  | Aggregate Fixed Income     | 2026-05-29 | 2    | -0.4% | 0.12 | 0.54  |
| BSV  | cluster_6  | Aggregate Fixed Income     | 2026-05-29 | 2    | -0.4% | 0.30 | 0.16  |
| SCHO | cluster_6  | Aggregate Fixed Income     | 2026-05-27 | 4    | -0.3% | 0.44 | 0.14  |
| AGZ  | cluster_6  | Aggregate Fixed Income     | 2026-05-11 | 15   | -0.3% | 0.24 | -0.05 |
| SHY  | cluster_6  | Aggregate Fixed Income     | 2026-05-27 | 4    | -0.3% | 0.52 | -0.06 |
| SPTS | cluster_6  | Aggregate Fixed Income     | 2026-05-27 | 4    | -0.2% | 0.55 | -0.11 |
| VGSH | cluster_6  | Aggregate Fixed Income     | 2026-05-27 | 4    | -0.3% | 0.56 | -0.12 |
| CMBS | cluster_6  | Aggregate Fixed Income     | 2026-05-29 | 2    | -0.7% | 0.64 | -0.26 |
| LQDH | cluster_23 | Bank Loan / Floating Ra... | 2026-04-06 | 40   | 1.6%  | 0.00 | 1.00  |
| IGBH | cluster_23 | Bank Loan / Floating Ra... | 2026-04-08 | 38   | 2.2%  | 0.00 | 1.00  |
| IGHG | cluster_23 | Bank Loan / Floating Ra... | 2026-04-07 | 39   | 1.2%  | 0.03 | 0.97  |
| BKLN | cluster_23 | Bank Loan / Floating Ra... | 2026-06-01 | 1    | 0.0%  | 1.00 | -0.95 |
| MINT | cluster_24 | Cash / Ultra-Short Fixe... | 2023-01-30 | 837  | 18.7% | 0.29 | 0.71  |
| BIL  | cluster_24 | Cash / Ultra-Short Fixe... | 2022-05-19 | 1011 | 17.9% | 0.44 | 0.56  |
| GBIL | cluster_24 | Cash / Ultra-Short Fixe... | 2022-08-18 | 949  | 17.4% | 0.52 | 0.48  |
| VNLA | cluster_24 | Cash / Ultra-Short Fixe... | 2026-04-02 | 41   | 0.3%  | 0.85 | 0.15  |
| IAI  | cluster_14 | Financials                 | 2026-04-10 | 36   | 7.2%  | 0.00 | 1.00  |
| IAT  | cluster_14 | Financials                 | 2026-05-20 | 8    | 0.8%  | 0.33 | 0.23  |
| KCE  | cluster_14 | Financials                 | 2026-04-13 | 35   | 2.6%  | 0.98 | 0.02  |
| XLF  | cluster_14 | Financials                 | 2026-04-13 | 35   | -0.4% | 0.66 | -0.02 |
| FNCL | cluster_14 | Financials                 | 2026-04-13 | 35   | -0.3% | 0.67 | -0.07 |
| PID  | cluster_8  | Asia-Pacific Equity        | 2026-04-30 | 22   | 1.7%  | 0.00 | 1.00  |
| EWC  | cluster_8  | Asia-Pacific Equity        | 2026-04-09 | 37   | 5.4%  | 0.01 | 0.99  |
| ENZL | cluster_8  | Asia-Pacific Equity        | 2026-05-22 | 6    | 2.8%  | 0.01 | 0.99  |
| EWL  | cluster_8  | Asia-Pacific Equity        | 2026-04-15 | 33   | 0.2%  | 0.03 | 0.97  |
| NFRA | cluster_8  | Asia-Pacific Equity        | 2026-04-09 | 37   | 1.6%  | 0.06 | 0.94  |
| EMLC | cluster_8  | Asia-Pacific Equity        | 2026-05-27 | 4    | 0.3%  | 0.06 | 0.88  |
| EBND | cluster_8  | Asia-Pacific Equity        | 2026-05-28 | 3    | -0.3% | 0.12 | 0.73  |
| ECH  | cluster_8  | Asia-Pacific Equity        | 2026-05-28 | 3    | -3.3% | 0.02 | 0.71  |
| EWV  | cluster_8  | Asia-Pacific Equity        | 2026-05-05 | 19   | 2.0%  | 0.30 | 0.64  |
| DWX  | cluster_8  | Asia-Pacific Equity        | 2026-05-04 | 20   | 0.9%  | 0.50 | 0.46  |
| EWA  | cluster_8  | Asia-Pacific Equity        | 2026-05-29 | 2    | 0.8%  | 0.31 | 0.40  |
| NURE | cluster_15 | Real Estate                | 2026-04-13 | 35   | 7.0%  | 0.06 | 0.94  |
| PSR  | cluster_15 | Real Estate                | 2026-04-09 | 37   | 3.0%  | 0.84 | 0.16  |
| SCHH | cluster_15 | Real Estate                | 2026-04-09 | 37   | 2.3%  | 0.86 | 0.14  |
| ICF  | cluster_15 | Real Estate                | 2026-04-08 | 38   | 3.4%  | 0.87 | 0.12  |
| USRT | cluster_15 | Real Estate                | 2026-04-09 | 37   | 3.1%  | 0.88 | 0.12  |
| REET | cluster_15 | Real Estate                | 2026-04-13 | 35   | 1.3%  | 0.88 | 0.11  |
| RWR  | cluster_15 | Real Estate                | 2026-04-10 | 36   | 2.2%  | 0.89 | 0.10  |
| IGOV | cluster_25 | International Fixed Inc... | 2026-05-27 | 4    | 0.3%  | 0.16 | 0.76  |
| FXB  | cluster_25 | International Fixed Inc... | 2026-05-22 | 6    | 0.1%  | 0.84 | -0.50 |
| LTPZ | cluster_21 | Inflation-Protected Fix... | 2026-05-28 | 3    | -1.0% | 0.06 | 0.85  |

|      |            |                            |            |    |       |      |       |
|------|------------|----------------------------|------------|----|-------|------|-------|
| SCHP | cluster_21 | Inflation-Protected Fix... | 2026-05-27 | 4  | -0.4% | 0.56 | 0.21  |
| SPIP | cluster_21 | Inflation-Protected Fix... | 2026-05-27 | 4  | -0.9% | 0.78 | -0.31 |
| TIP  | cluster_21 | Inflation-Protected Fix... | 2026-05-28 | 3  | -1.1% | 0.85 | -0.45 |
| TDTF | cluster_21 | Inflation-Protected Fix... | 2026-05-28 | 3  | -1.2% | 1.00 | -0.95 |
| SILJ | cluster_17 | Gold                       | 2026-06-02 | 0  | 0.0%  | 0.48 | 0.22  |
| SIVR | cluster_17 | Gold                       | 2026-05-27 | 4  | 0.7%  | 0.85 | -0.34 |
| SLV  | cluster_17 | Gold                       | 2026-05-27 | 4  | 0.7%  | 0.85 | -0.34 |
| CNXT | cluster_12 | China Equity               | 2026-04-10 | 36 | 19.3% | 0.01 | 0.99  |
| CQQQ | cluster_12 | China Equity               | 2026-04-20 | 30 | 6.7%  | 0.63 | 0.37  |
| UUP  | cluster_26 | Leveraged / Inverse        | 2026-05-18 | 10 | 0.2%  | 0.03 | 0.96  |

## ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

### Top 20 most bullish (cluster-deduped, abs-mom approved)

| ticker | cluster    | theme                        | p_buy | nss  |
|--------|------------|------------------------------|-------|------|
| HACK   | cluster_5  | Global Equity                | 1.00  | 1.00 |
| QQEW   | cluster_1  | U.S. Large Cap Equity        | 1.00  | 1.00 |
| FDVV   | cluster_3  | U.S. Large Cap Value         | 1.00  | 1.00 |
| XTN    | cluster_2  | U.S. Mid Cap Equity          | 1.00  | 1.00 |
| ARKG   | cluster_11 | Healthcare                   | 1.00  | 1.00 |
| LQDH   | cluster_23 | Bank Loan / Floating Rate    | 1.00  | 1.00 |
| IHDG   | cluster_4  | Developed ex-US Equity       | 1.00  | 1.00 |
| IAI    | cluster_14 | Financials                   | 1.00  | 1.00 |
| EEMV   | cluster_7  | Emerging Markets Equity      | 1.00  | 1.00 |
| IAGG   | cluster_6  | Aggregate Fixed Income       | 1.00  | 1.00 |
| PID    | cluster_8  | Asia-Pacific Equity          | 1.00  | 1.00 |
| DBJP   | cluster_22 | Japan Equity                 | 1.00  | 1.00 |
| SHM    | cluster_16 | Municipal Bond               | 1.00  | 1.00 |
| SLX    | cluster_9  | Materials                    | 1.00  | 0.99 |
| CNXT   | cluster_12 | China Equity                 | 1.00  | 0.99 |
| SPLB   | cluster_13 | Fixed Income                 | 0.99  | 0.99 |
| EMB    | cluster_10 | High Yield Bond              | 1.00  | 0.98 |
| NURE   | cluster_15 | Real Estate                  | 1.00  | 0.94 |
| LTPZ   | cluster_21 | Inflation-Protected Fixed... | 0.91  | 0.85 |
| MINT   | cluster_24 | Cash / Ultra-Short Fixed ... | 1.00  | 0.71 |

### Top 20 most bearish (cluster-deduped)

| ticker | cluster    | theme                 | p_sell | nss   |
|--------|------------|-----------------------|--------|-------|
| EIDO   | cluster_19 | India Equity          | 1.00   | -1.00 |
| FXC    | cluster_8  | Asia-Pacific Equity   | 1.00   | -1.00 |
| CHIQ   | cluster_12 | China Equity          | 1.00   | -1.00 |
| RTH    | cluster_1  | U.S. Large Cap Equity | 1.00   | -1.00 |
| DBA    | cluster_20 | Broad Commodities     | 1.00   | -1.00 |
| SQQQ   | cluster_26 | Leveraged / Inverse   | 1.00   | -1.00 |

|      |            |                              |      |       |
|------|------------|------------------------------|------|-------|
| MOO  | cluster_9  | Materials                    | 1.00 | -1.00 |
| SDIV | cluster_7  | Emerging Markets Equity      | 1.00 | -1.00 |
| PBJ  | cluster_18 | Consumer Staples             | 0.98 | -0.98 |
| SPLV | cluster_3  | U.S. Large Cap Value         | 0.98 | -0.98 |
| TDTT | cluster_21 | Inflation-Protected Fixed... | 1.00 | -0.97 |
| KIE  | cluster_14 | Financials                   | 0.98 | -0.97 |
| PGF  | cluster_10 | High Yield Bond              | 0.97 | -0.97 |
| BKLN | cluster_23 | Bank Loan / Floating Rate    | 1.00 | -0.95 |
| FXY  | cluster_25 | International Fixed Income   | 1.00 | -0.95 |
| GLD  | cluster_17 | Gold                         | 0.95 | -0.93 |
| HDEF | cluster_4  | Developed ex-US Equity       | 0.93 | -0.73 |
| IHI  | cluster_11 | Healthcare                   | 0.54 | -0.54 |
| ESPO | cluster_5  | Global Equity                | 0.44 | -0.42 |
| SDVY | cluster_2  | U.S. Mid Cap Equity          | 0.72 | -0.35 |

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## Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

| ticker | name                               | category                 | cluster_id | cluster_size | etf_signal | max_corr_to_others | median_corr_to_others | abs_pc1_loading | compo |
|--------|------------------------------------|--------------------------|------------|--------------|------------|--------------------|-----------------------|-----------------|-------|
| PPLT   | abrdn Physical Platinum Shares ... | Platinum                 | cluster_19 | 11           | FLAT       | 0.302              | 0.093                 | 0.00021         |       |
| UNG    | United States Natural Gas Fund ... | Natural Gas              | cluster_19 | 11           | FLAT       | 0.490              | 0.077                 | 0.00017         |       |
| GBIL   | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.707              | 0.040                 | 0.00007         |       |
| FXV    | Invesco CurrencyShares Japanese... | Currency                 | cluster_25 | 6            | FLAT       | 0.731              | 0.094                 | 0.00021         |       |
| VNLA   | Janus Henderson Short Duration ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.615              | 0.219                 | 0.00035         |       |
| FTSM   | First Trust Enhanced Short Matu... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.762              | 0.133                 | 0.00030         |       |
| DBA    | Invesco DB Agriculture Fund        | Agricultural Commodit... | cluster_20 | 10           | FLAT       | 0.596              | 0.168                 | 0.00040         |       |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.762              | 0.161                 | 0.00038         |       |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.842              | 0.024                 | 0.00000         |       |
| MINT   | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.656              | 0.188                 | 0.00046         |       |
| NEAR   | iShares Short Duration Bond Act... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.707              | 0.189                 | 0.00043         |       |
| SHV    | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.842              | 0.084                 | 0.00021         |       |
| GSY    | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.756              | 0.180                 | 0.00043         |       |
| TUR    | iShares MSCI Turkey ETF            | Turkey Equity            | cluster_19 | 11           | FLAT       | 0.498              | 0.216                 | 0.00055         |       |
| LDUR   | PIMCO Enhanced Low Duration Act... | Fixed Income             | cluster_6  | 43           | FLAT       | 0.845              | 0.126                 | 0.00034         |       |
| FXF    | Invesco CurrencyShares Swiss Fr... | Currency                 | cluster_25 | 6            | FLAT       | 0.818              | 0.202                 | 0.00048         |       |
| CMBS   | iShares CMBS Bond ETF              | Mortgage-Backed Fixed... | cluster_6  | 43           | LONG       | 0.877              | 0.161                 | 0.00030         |       |
| LMBS   | First Trust Low Duration Opport... | Mortgage-Backed Fixed... | cluster_6  | 43           | FLAT       | 0.836              | 0.236                 | 0.00046         |       |
| NOBL   | ProShares S&P 500 Dividend Aris... | U.S. Dividend Equity     | cluster_19 | 11           | FLAT       | 0.445              | 0.277                 | 0.00075         |       |
| SHM    | State Street SPDR Nuveen ICE Sh... | Municipal Bond           | cluster_16 | 13           | LONG       | 0.878              | 0.250                 | 0.00045         |       |
| VNM    | VanEck Vietnam ETF                 | Vietnam Equity           | cluster_19 | 11           | FLAT       | 0.445              | 0.330                 | 0.00083         |       |
| SUB    | iShares Short-Term National Mun... | Municipal Bond           | cluster_16 | 13           | LONG       | 0.878              | 0.265                 | 0.00052         |       |
| KURE   | KraneShares MSCI All China Heal... | China Equity             | cluster_12 | 15           | FLAT       | 0.805              | 0.269                 | 0.00071         |       |
| DBB    | Invesco DB Base Metals Fund        | Industrial Metals        | cluster_9  | 21           | LONG       | 0.750              | 0.359                 | 0.00084         |       |
| EIDO   | iShares MSCI Indonesia ETF         | Indonesia Equity         | cluster_19 | 11           | FLAT       | 0.500              | 0.367                 | 0.00105         |       |
| KBA    | KraneShares Bosera MSCI China A... | China Equity             | cluster_12 | 15           | FLAT       | 0.855              | 0.255                 | 0.00071         |       |
| EPHE   | iShares MSCI Philippines ETF       | Philippines Equity       | cluster_19 | 11           | FLAT       | 0.571              | 0.411                 | 0.00107         |       |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed I... | cluster_23 | 9            | FLAT       | 0.866              | 0.329                 | 0.00071         |       |
| LTPZ   | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected F... | cluster_21 | 10           | LONG       | 0.918              | 0.226                 | 0.00056         |       |
| CNXT   | VanEck ChiNext Innovators ETF      | China Equity             | cluster_12 | 15           | LONG       | 0.904              | 0.241                 | 0.00063         |       |

## Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

|                          | scope  | parameter                | value |
|--------------------------|--------|--------------------------|-------|
| short_ma                 | screen | short_ma                 | 5     |
| buy_ma                   | screen | buy_ma                   | 50    |
| sell_ma                  | screen | sell_ma                  | 20    |
| vol_window               | screen | vol_window               | 20    |
| horizon                  | screen | horizon                  | 5     |
| imminent_horizon         | screen | imminent_horizon         | 1     |
| cluster_smoothing_window | screen | cluster_smoothing_window | 3     |
| sell_cushion_k           | screen | sell_cushion_k           | 0.5   |
| sell_cushion_sd_window   | screen | sell_cushion_sd_window   | 20    |